

Orbital Mechanics

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Topics

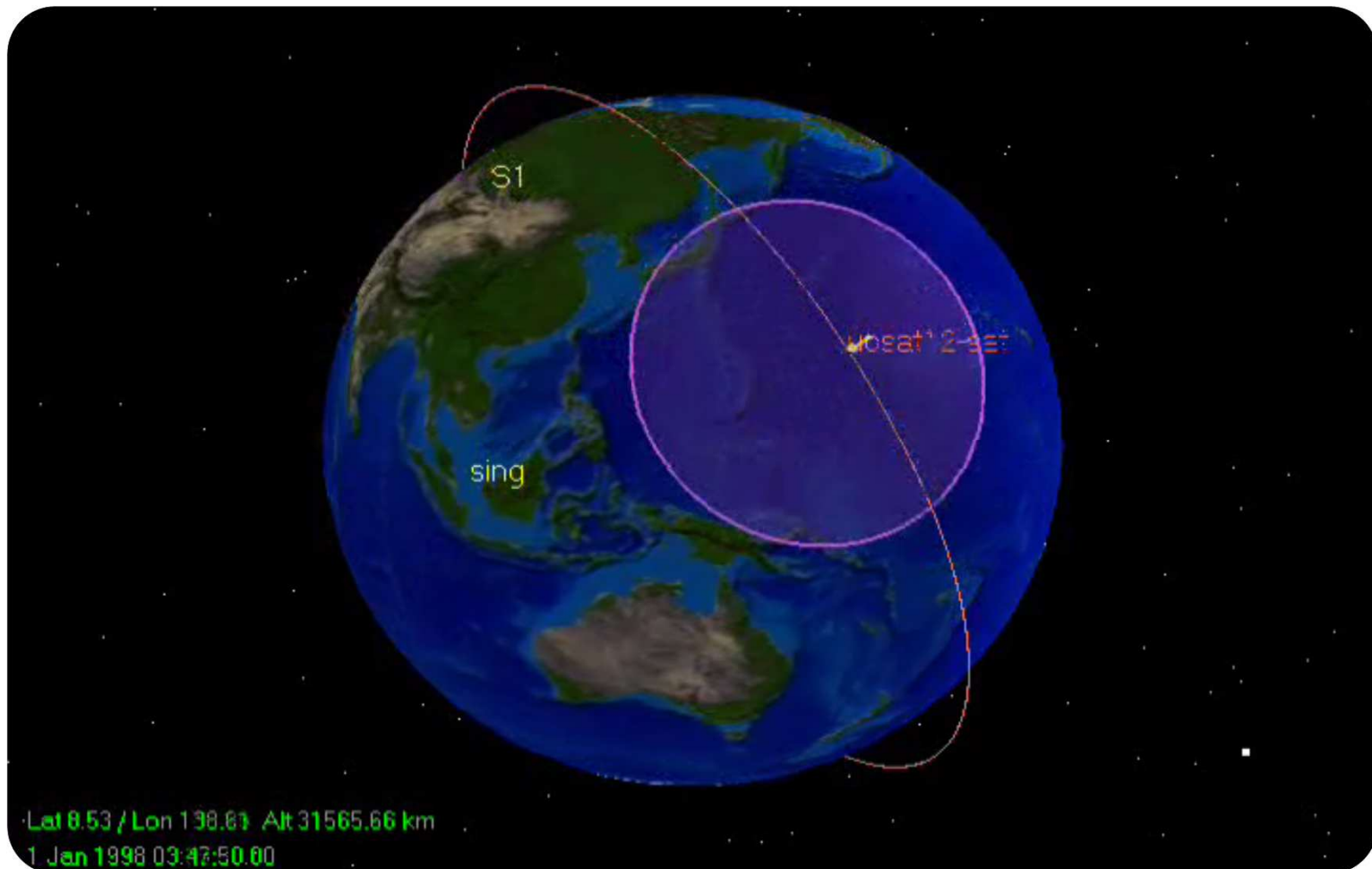
- Basics of Orbit
- Classification of orbits
- Definition of orbital elements
- Coordinate Systems
- Coordinate Transformations
- Basic equations of orbits
- Orbit perturbations
- Orbit Maneuvers

- Orbit Propagation
- Two Line Elements
- Orbit Determination
- Events
 - Ground Traces
 - Coverage Calculations
 - Orbital Eclipses
- Constellations Design

Typical Low Earth Orbit (LEO)

Altitude 700 kms

Orbit Period ~ 100 min



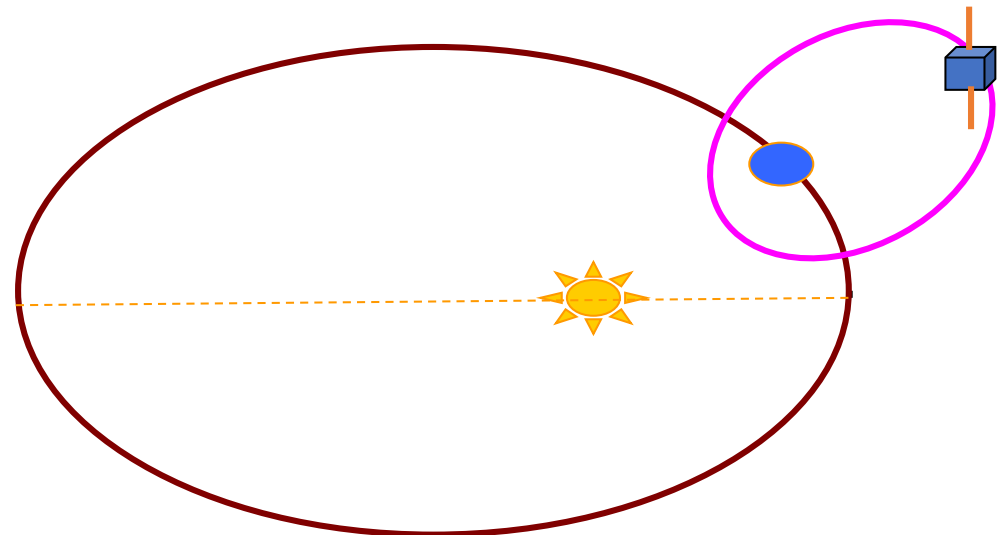
Kepler Laws (1/3)

- Kepler laws – are based on observations and curve fits.
- First Law: Planets move around Sun in Elliptical Orbit planes.
 - *(In case of satellites natural or man made, the satellites move around the planet, say Earth, in elliptical orbit with the planet at one of its focal points)*

Kepler



1571- 1630

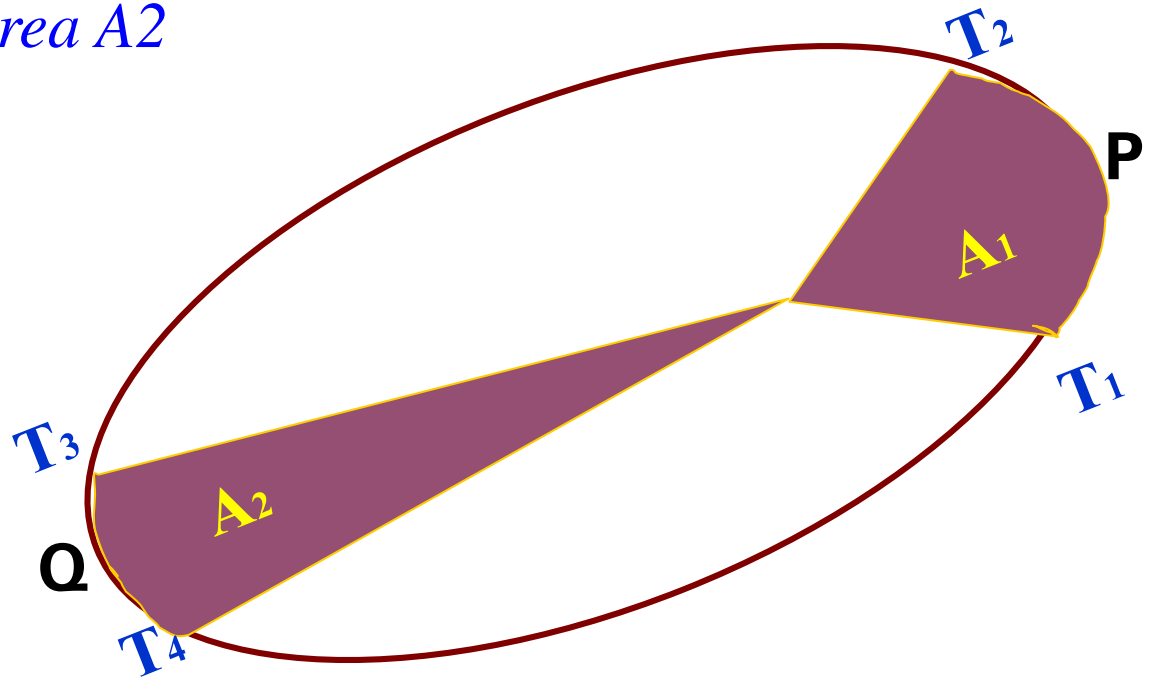


Ref: [Kepler's Laws - MacTutor History of Mathematics](#)

Kepler Laws (2/3)

- Second Law: Planet sweeps equal areas in equal time intervals.
- *This means,*
when $T_2 - T_1 = T_4 - T_3$,
Area $A_1 = \text{Area } A_2$

What does it imply to the velocity at P and Q?



Kepler Laws (3/3)

- Third Law: Square of Period (T) is proportional to cube of the semi-major axis (a)

$$T^2 = 4\pi^2 \left(\frac{a^3}{\mu} \right)$$

$$a = R_E + h ;$$

$$R_E = 6378 \text{ Kms}$$

$$\mu = 398600.64 \text{ km}^3/\text{s}^2$$

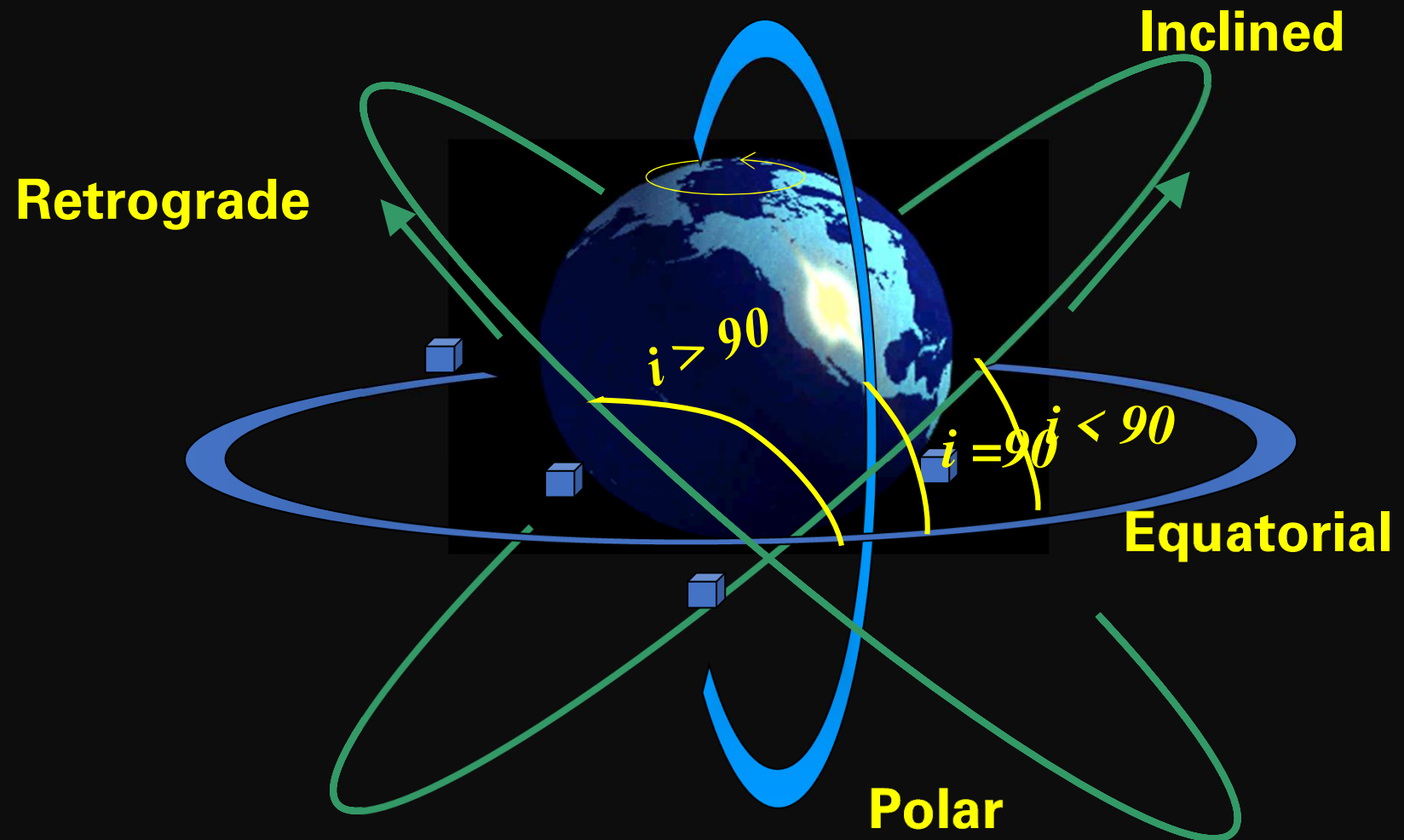
Height (h)	Period (T)	Application
750	100 min	Remote sensing
10380	6 hr	
20230	12 hr	GPS_Navigation
35785	24 hr	GEO-Comm.sat

Classification of Orbits

Orbits can be classified based on,
Inclination
Altitude
Shape
Application

TYPES OF ORBITS

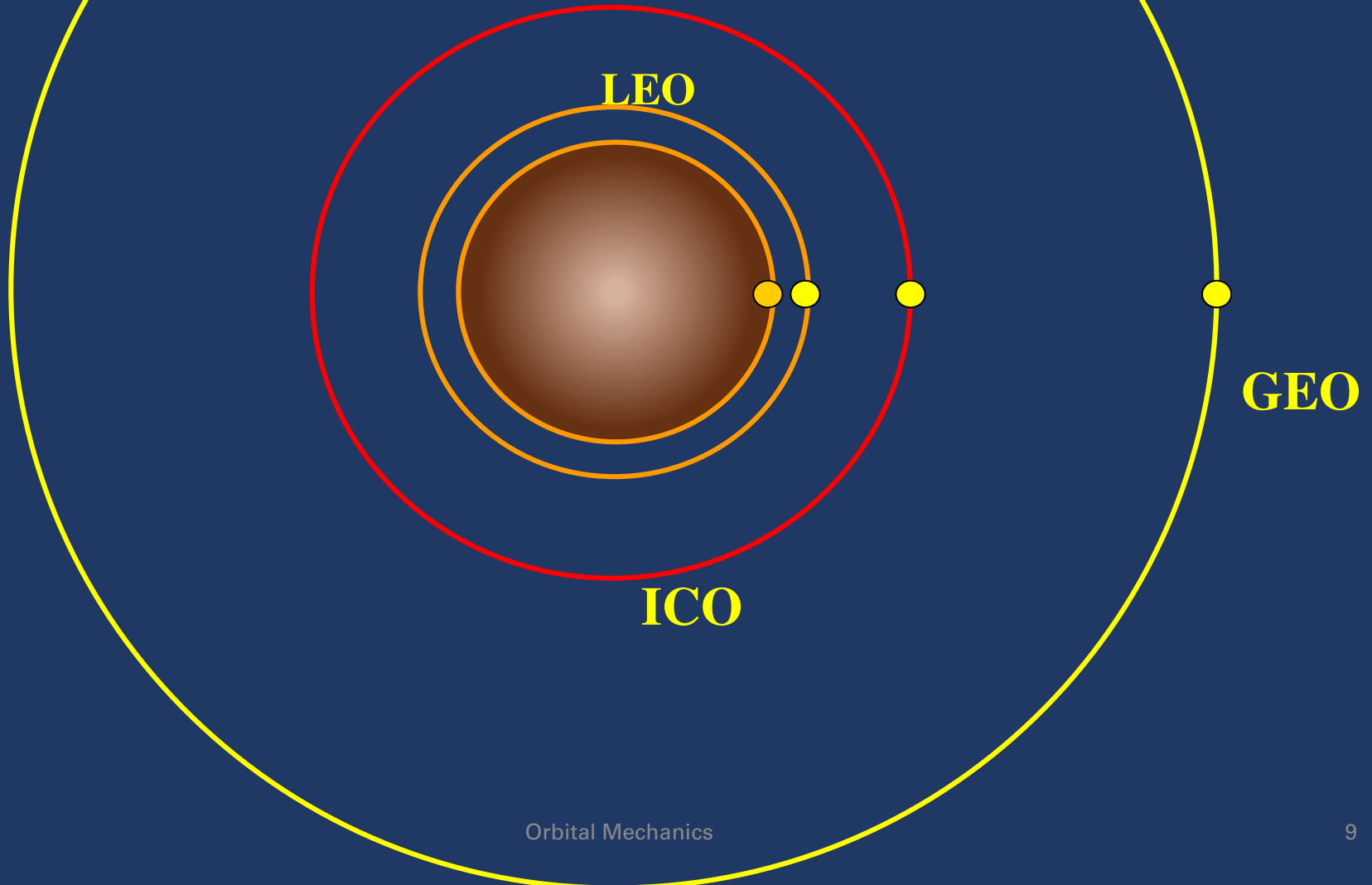
Based on Inclination



TYPES OF ORBITS

(Based on altitude)

$$T^2 = 4\pi^2 \left(\frac{a^3}{\mu} \right)$$



TYPES OF ORBITS (Summary)

Inclination

Equatorial
Inclined
Polar

Altitude

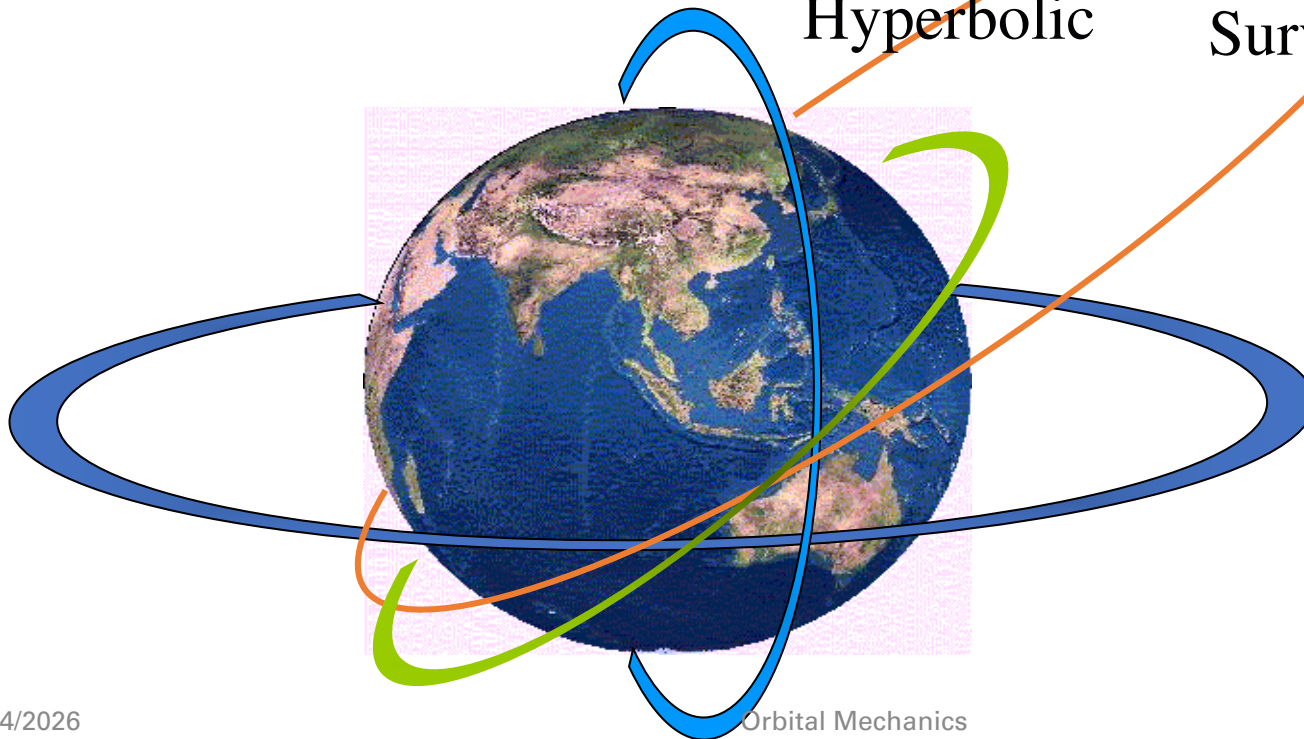
LEO
ICO
GSO

Shape

Circular
Elliptic
Parabolic
Hyperbolic

Application

Remote sensing(SSO)
Communication
Navigation
Surveillance



LEO, ICO, GEO Missions

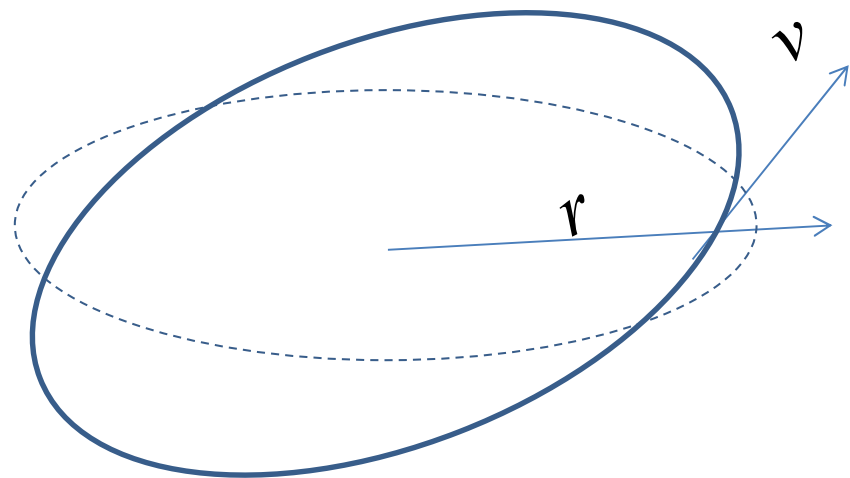
$$T^2 = 4\pi^2 \left(\frac{a^3}{\mu} \right)$$

Mission	Height	Period	Application	Satellites
LEO	750	100 min	Remote sensing (SSO)	Landsat, Spot, IRS
ICO	10380	6 hr	Communication	ICO (Boeing) (www.ico.com)
ICO	20230	12 hr	Navigation	GPS
GEO	35785	24 hr	GEO-Comm.sat	Intelsat, Insat, ATS

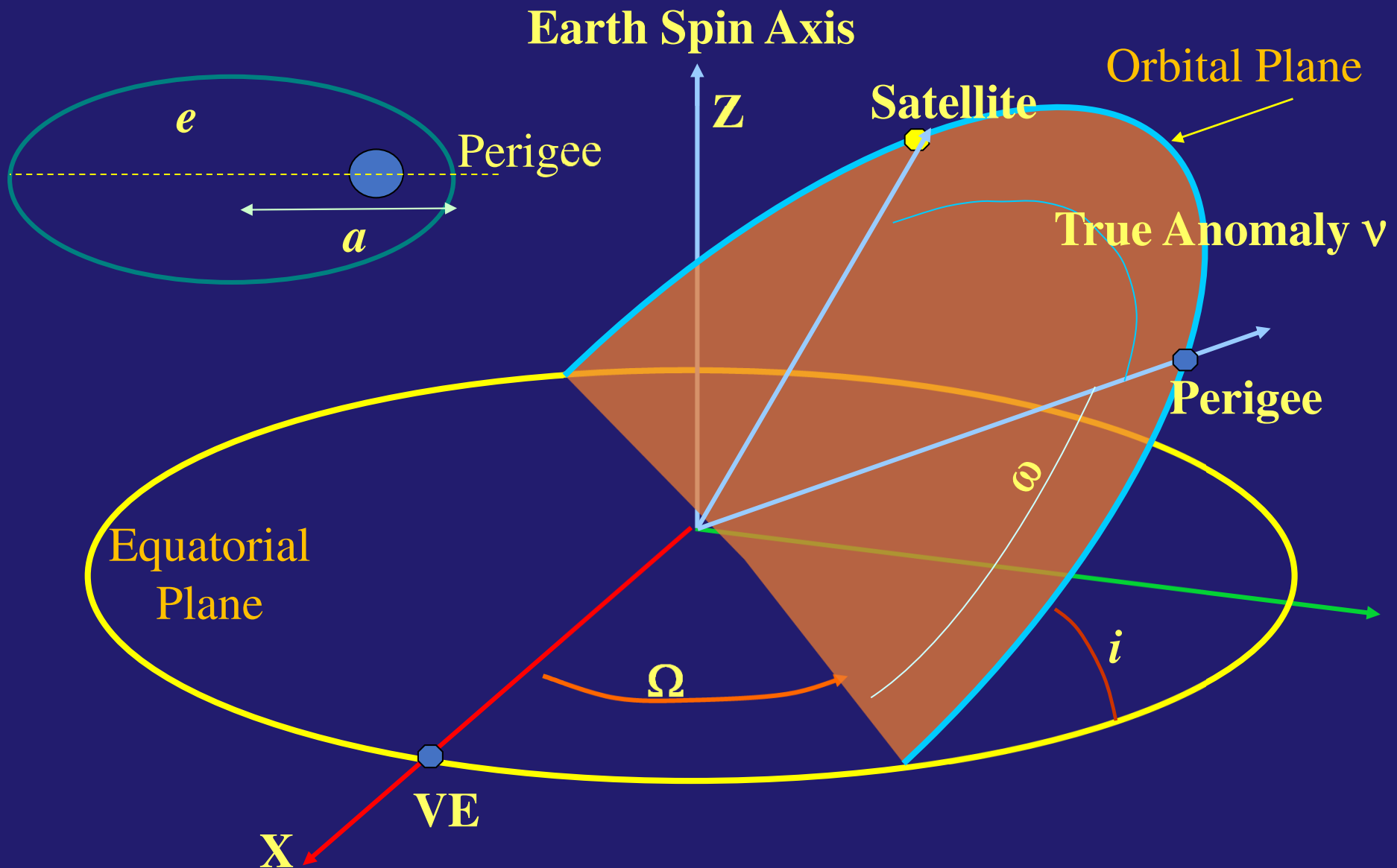
Orbital Elements

Orbital Elements

- The path of an orbit is completely specified once the Position and Velocity vectors at a time t are known.
- The path can be expressed in terms of Orbital Elements (commonly Kepler Elements) which give better insight



Orbital Elements



Kepler Elements

Symbol	Description	Remarks
a	Semi Major Axis	Determines the orbital period
e	Eccentricity	Determines the shape of orbit
i	Inclination	Determines how the orbital plane is tilted w.r.t the equatorial plane
ω	Argument of perigee	Determines the location of perigee-apogee within the orbital plane
Ω	RAAN	Determines the orientation of orbital plane in inertial space
M_0	Mean Anomaly	Determines the location of satellite in the orbit defined by the previous five elements
T_0	Epoch Time	Universal Time corresponding to M_0

Important Orbital Equations

Orbit Equations

Radius Vector r

$$r = \frac{a(1 - e^2)}{1 + e\cos\theta} = a(1 - e\cos E)$$

**Apogee radius (r_a),
Perigee radius (r_p)**

$$r_a = a(1 + e), \quad r_p = a(1 - e)$$

Velocity (v)

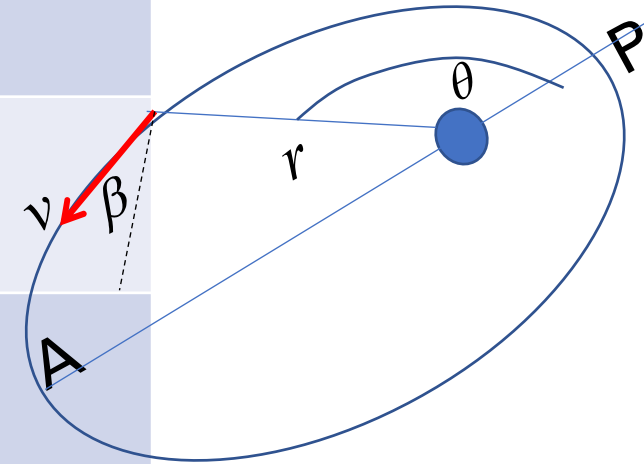
$$v^2 = \mu \left[\frac{2}{r} - \frac{1}{a} \right] \quad \text{red } v_c^2 = \left[\frac{\mu}{r} \right]$$

**Apogee velocity (v_a)
Perigee velocity (v_p)**

$$v_a^2 = \frac{\mu}{a} \left[\frac{1 - e}{1 + e} \right], \quad v_p^2 = \frac{\mu}{a} \left[\frac{1 + e}{1 - e} \right]$$

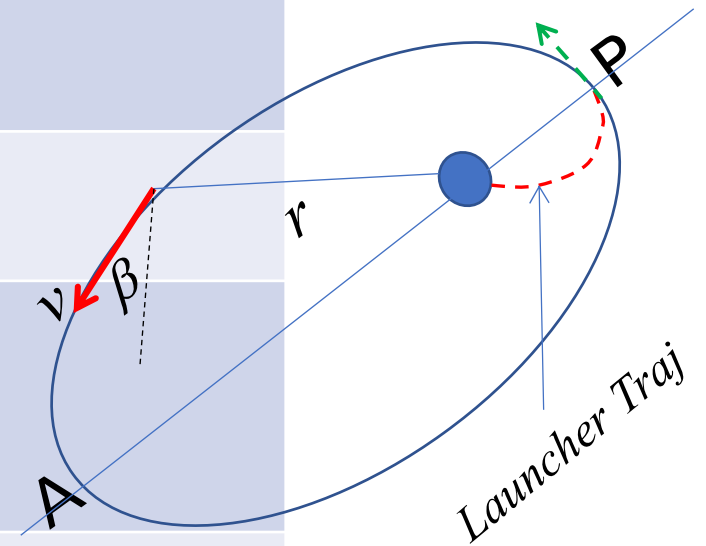
Flight Path Angle (β)

$$\beta = \tan^{-1} \left[\frac{e\sin\theta}{1 + e\cos\theta} \right]$$



Orbit Equations

Eccentricity e	$e = \frac{r_a - r_p}{2a}$
Semi minor axis b	$b = a\sqrt{1 - e^2}$
Mean Motion (η)	$\eta = \sqrt{\frac{\mu}{a^3}}$
Orbital Period (T)	$T = \frac{2\pi}{\eta} = \sqrt{\frac{a^3}{\mu}}$



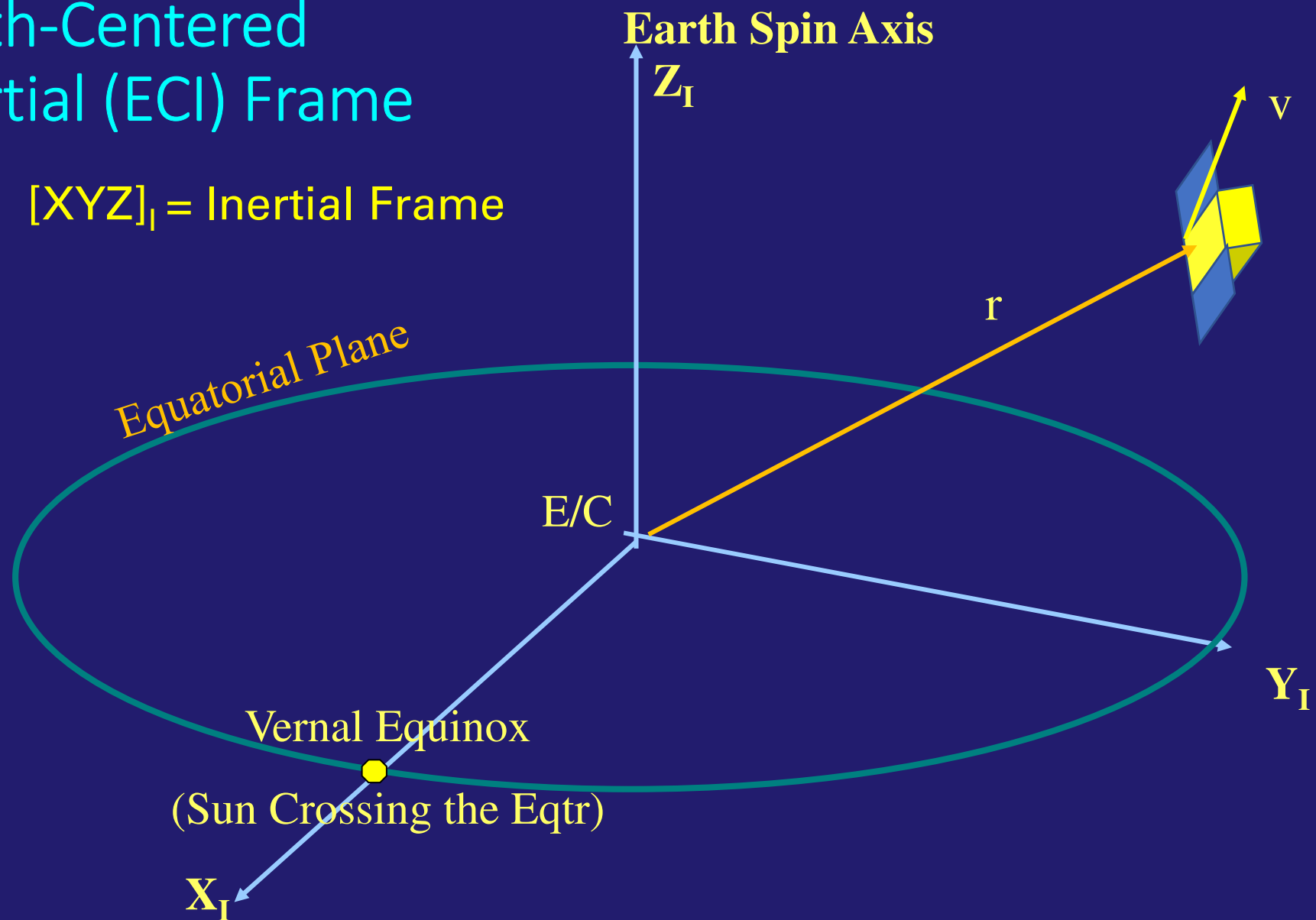
COORDINATE SYSTEMS AND ROTATIONS

Coordinate Systems and Rotations

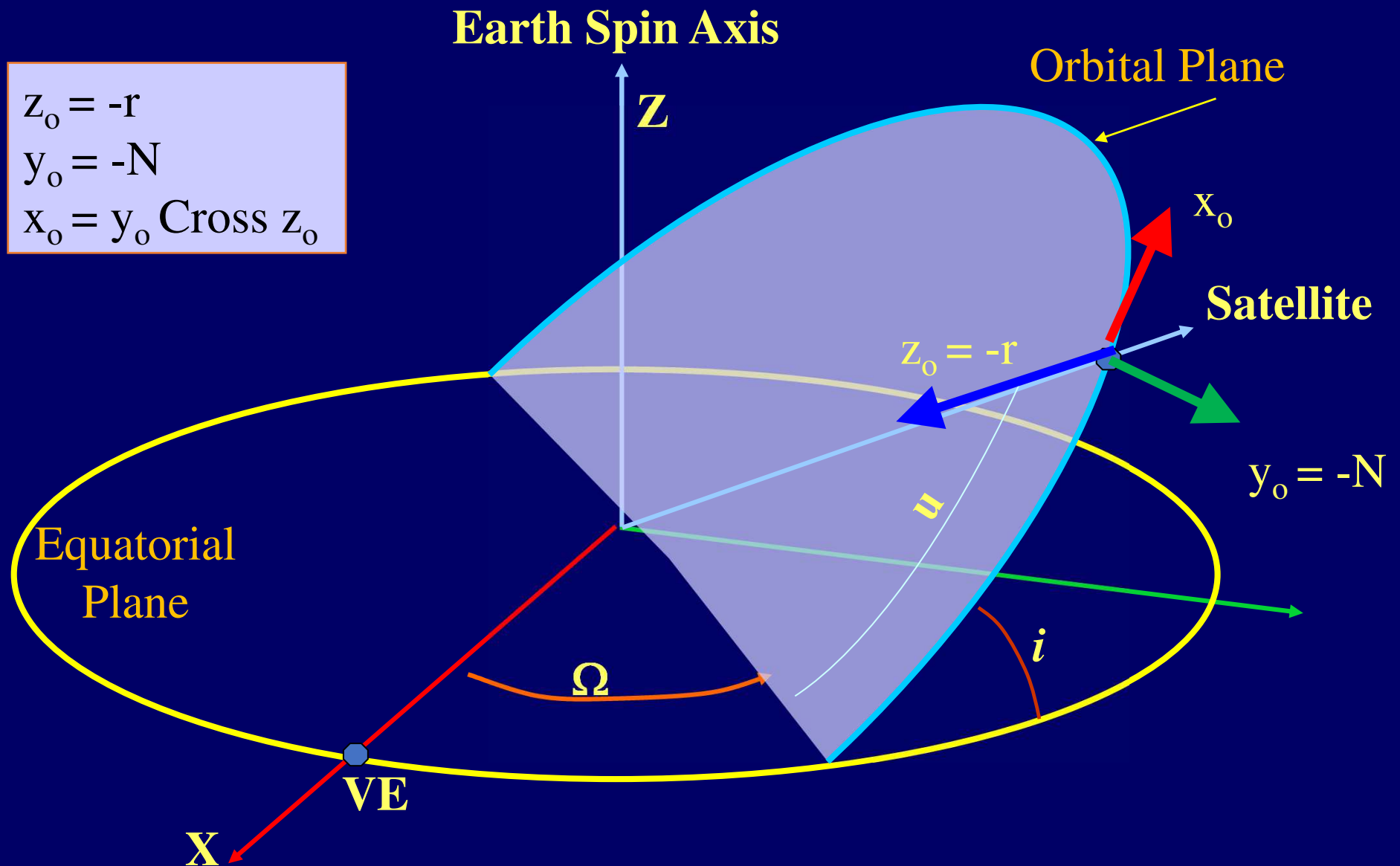
- We use different coordinate systems to represent vectors and angles according to the convenience
 - Sensors, actuators, modules are represented in **body coordinate** system in CAD model.
 - Ground station is represented by Lat, Long, Alt in **Geodetic coordinate** system.
 - Ground-to-target directions are measured in Aimuth, Elevation, range in **Topocentric system**.
 - Stars, Planets, Sun are calculated in **Inertial system**.
- Very often we transform vectors from one frame to another for mathematical calculations, simulations, analysis etc

Earth-Centered Inertial (ECI) Frame

$[XYZ]_I$ = Inertial Frame



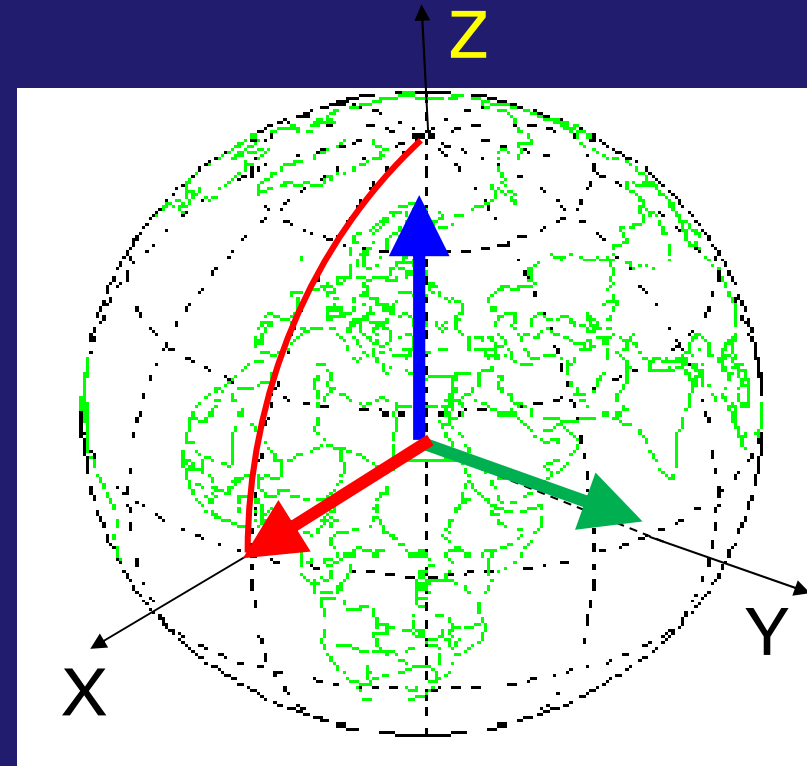
Orbital Reference System



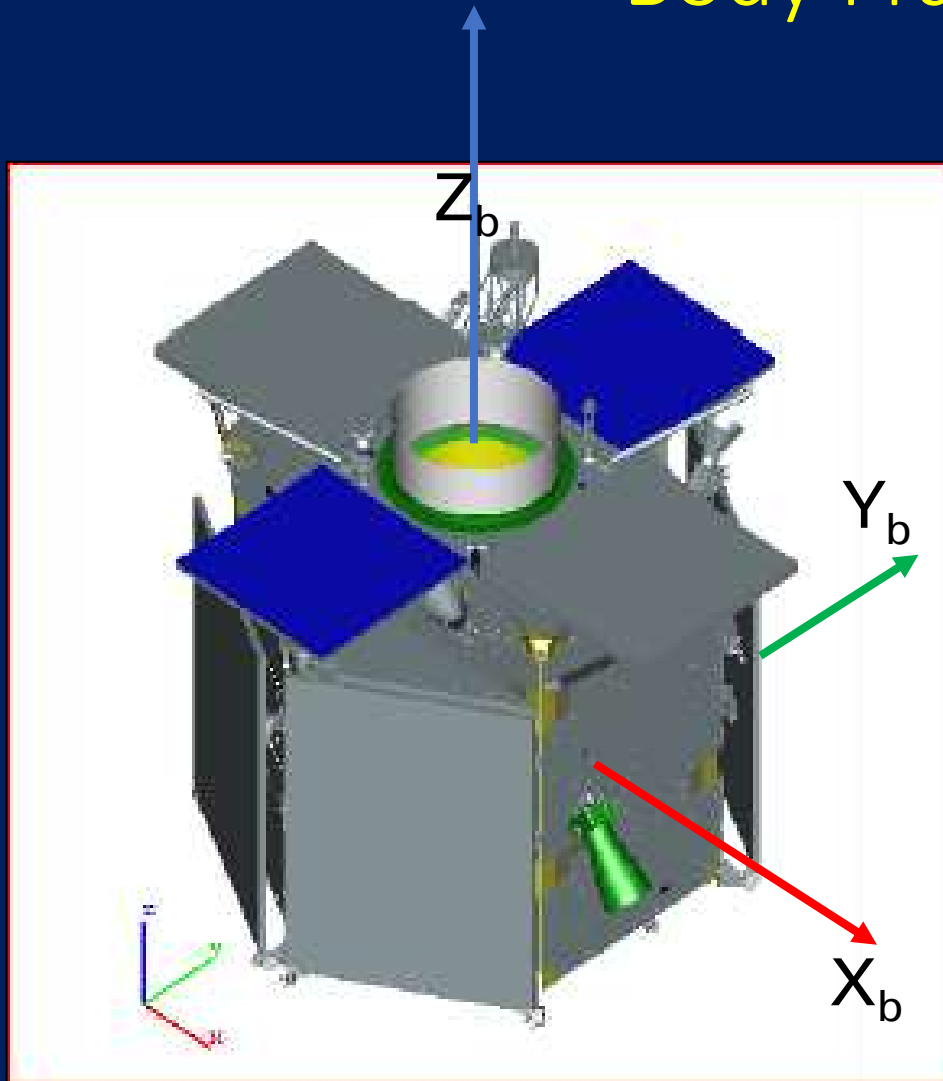
Earth Fixed Coordinate System

$[XYZ]_g$

- Z along spin axis
- X along the equator on the Greenwich meridian
- $Y = Z \times X$
- Useful to define the location of ground station, direction of geomagnetic field...
- X, Y, Z components are used to calculate the longitude, latitude and altitude.



Body Frame



- This is useful to define the mechanical configuration
- Mounting of sensors, actuators and other modules are specified in this system

Coordinate Rotations

- What for ?
 - Coordinate rotations are required to transform a vector from a given reference frame to another frame.
 - To transform the position and velocity vectors to Earth fixed system, topocentric for calculating geographic coordinates, antenna pointing directions for tracking.
 - During simulation, the Sun, Magnetic, Star Vectors need to be transformed from Inertial to body and sensor frame to get sensor measurements
- This may involve simple rotation about one axis or complex rotations involving a series of rotations about different axes.

Inertial to Earth Fixed Frame

$[XYZ]_I$ = Inertial Frame;

$[XYZ]_g$ = Earth Fixed Frame

Rotate by θ about Z_I axis

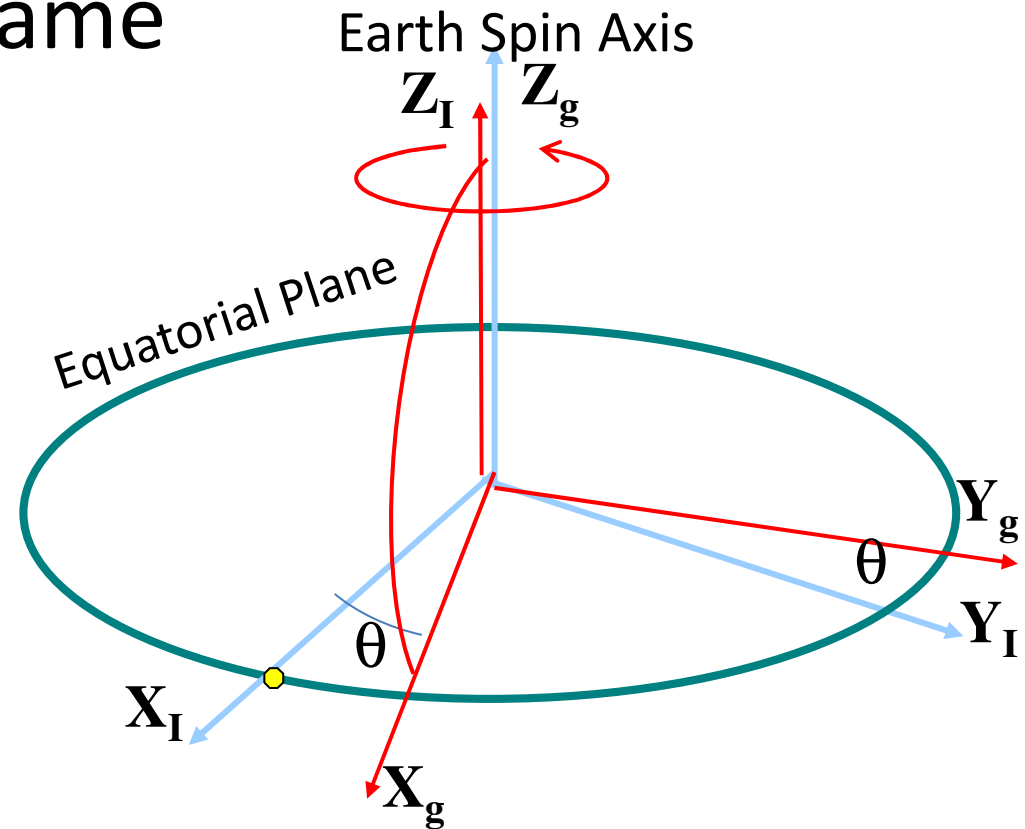
$$\begin{aligned} X_g &= \cos\theta \hat{i} + \sin\theta \hat{j} + 0 \hat{k} \\ Y_g &= -\sin\theta \hat{i} + \cos\theta \hat{j} + 0 \hat{k} \\ Z_g &= 1 \hat{k} \end{aligned} \quad \text{E/C}$$

In matrix form,

$$\begin{bmatrix} X_g \\ Y_g \\ Z_g \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_I \\ Y_I \\ Z_I \end{bmatrix} ;$$

$$X_g = A X_I$$

**A=Transformation
Matrix**



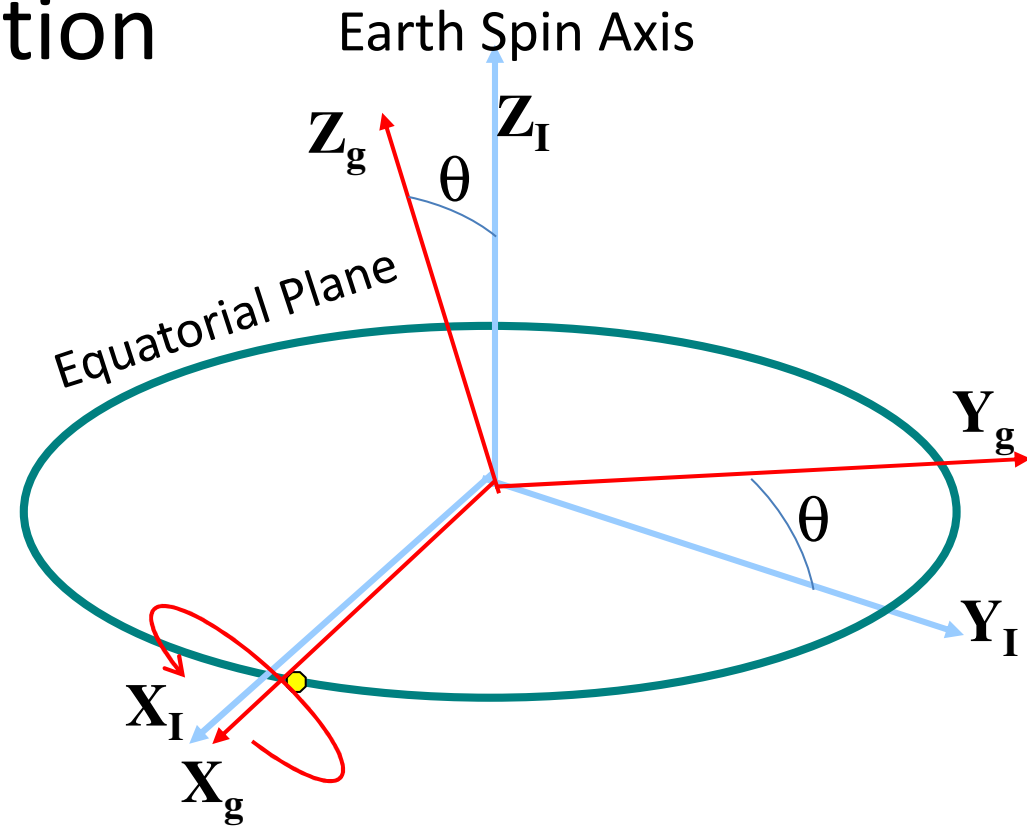
Another Single Axis Rotation

Rotate by θ about X_I axis

In matrix form,

$$\begin{bmatrix} X_g \\ Y_g \\ Z_g \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} X_I \\ Y_I \\ Z_I \end{bmatrix}$$

E/C



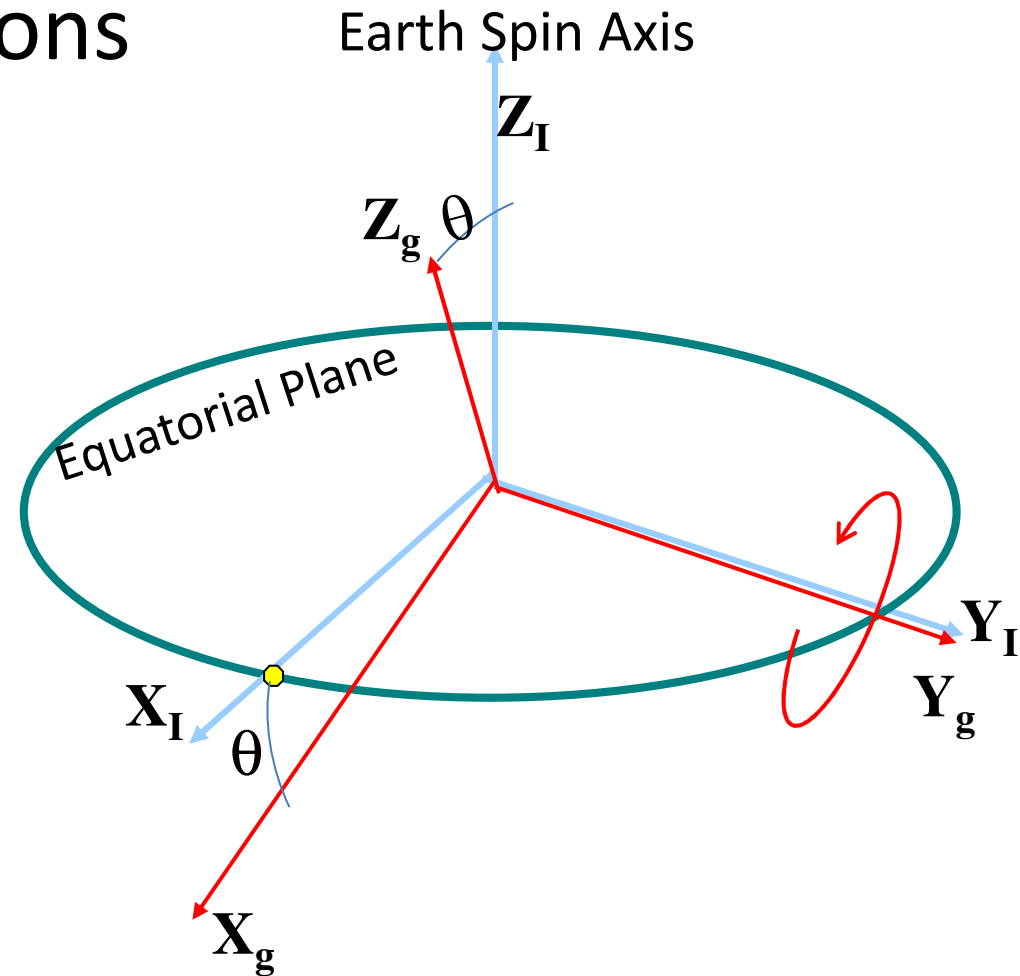
Other Single Axis Rotations

Rotate by θ about Y_I axis

In matrix form,

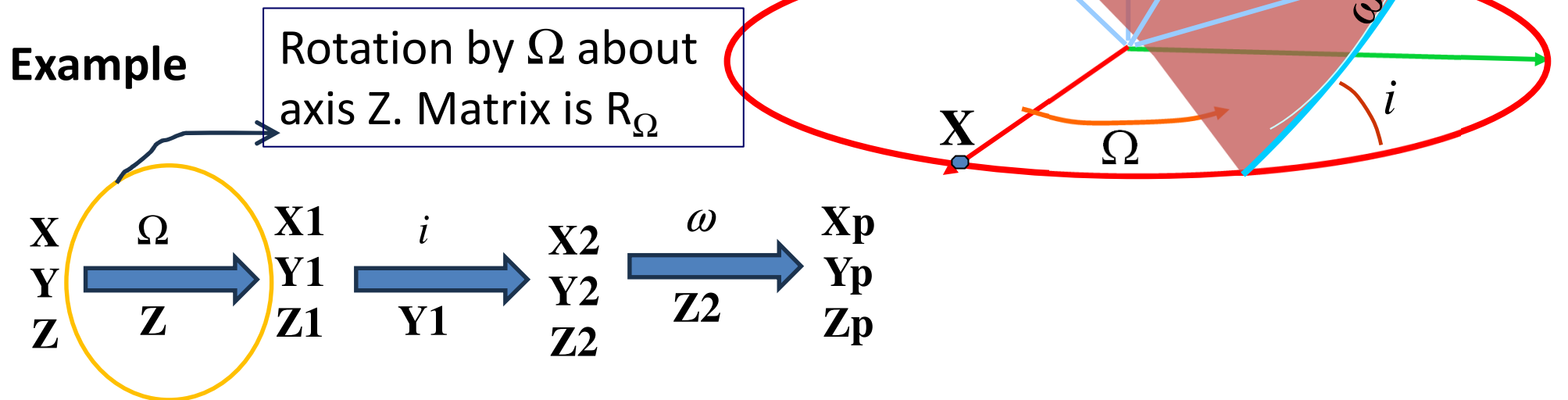
$$\begin{bmatrix} X_g \\ Y_g \\ Z_g \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} X_I \\ Y_I \\ Z_I \end{bmatrix}$$

E/C



Multiple Rotations

In any computations, we will encounter multiple rotations. But they can be handled as sequence of 'Single Rotations'



Final Rotation Matrix $T = R_{\omega} R_i R_{\Omega}$

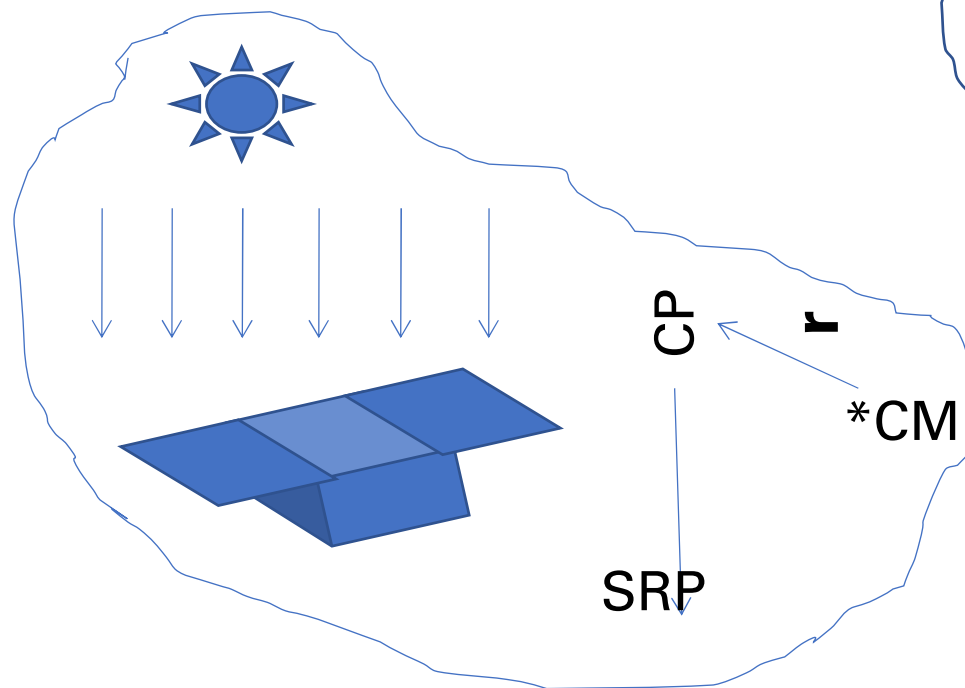
ORBITAL PERTURBATIONS

Orbital Perturbations

- Keplerian Orbital Parameters are valid only under two-body environment (central force)
- In reality, there are additional disturbing forces (though small) which act on the satellite all the times.
- These disturbing forces, perturb the orbit and hence the orbital parameters change with time.
- Perturbations to orbital parameters are – secular, long periodic and short periodic in nature.

Perturbation Sources (1/2)

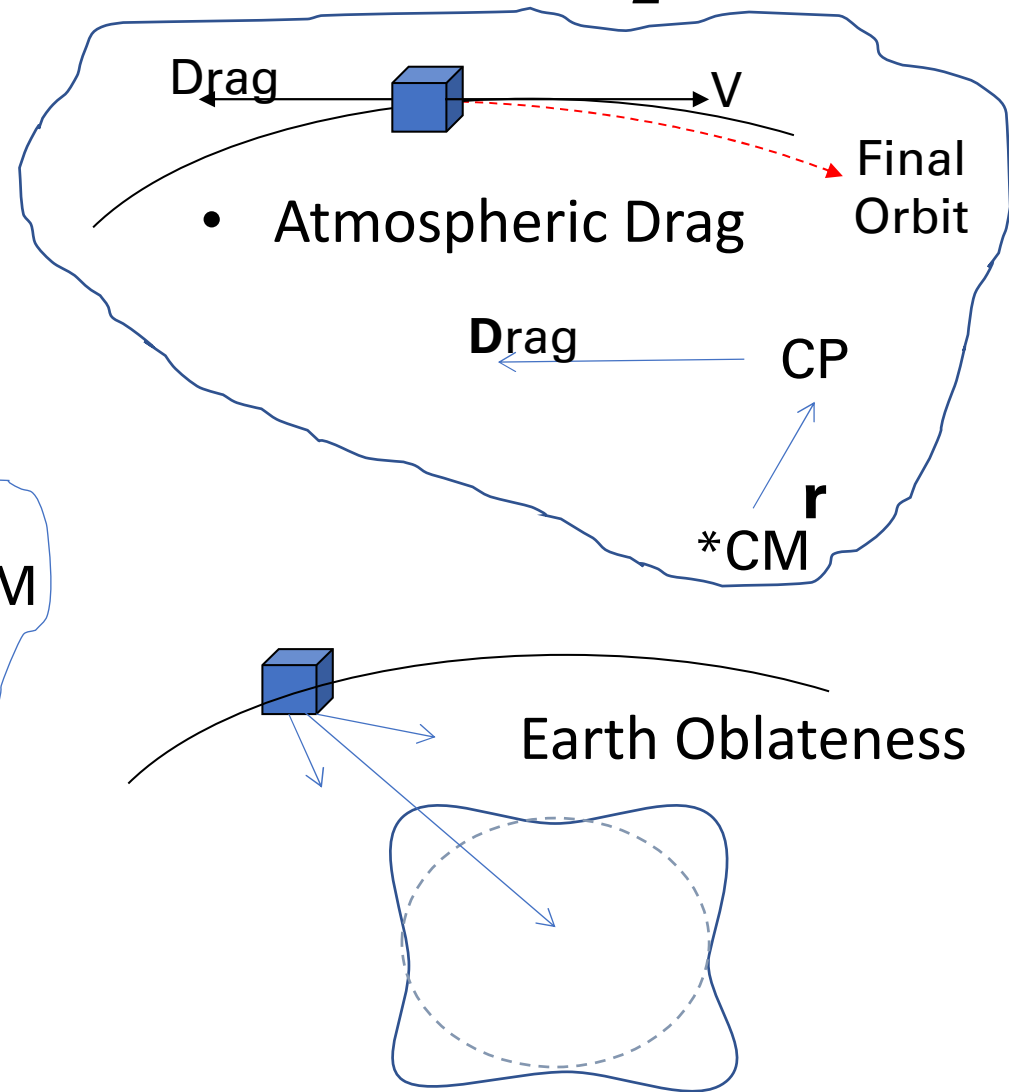
- Solar Radiation Pressure



$$F_{SRP} = -4.5 \times 10^{-6} (1 + r) A$$

$r = \text{Reflection factor } 0 < r < 1$

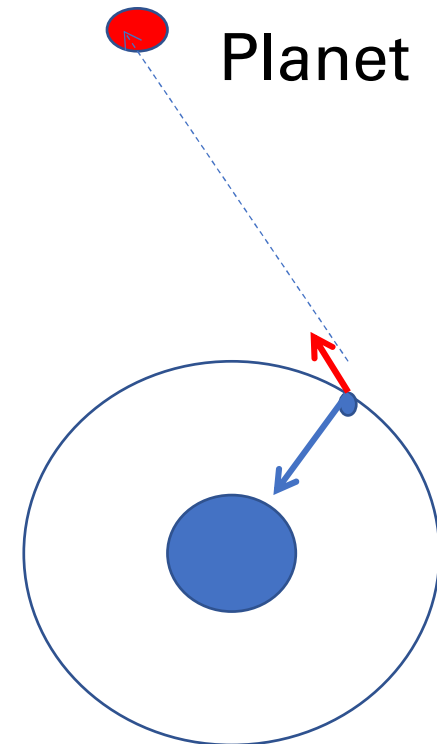
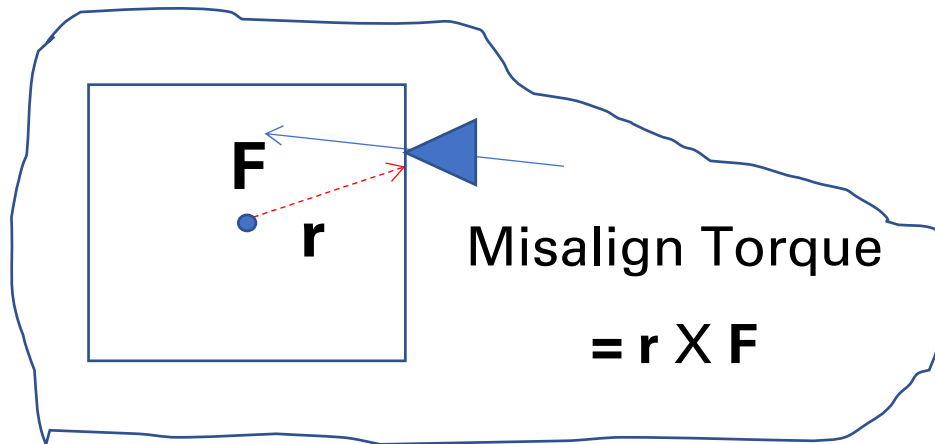
$$F_d = \frac{1}{2} \rho v^2 C_d A$$



Perturbation Sources (2/2)

Thruster Misalignments

- Thruster mounting error
- Thrust vector error during firing



Third Body attractions

Case Study: 3U CubeSat

Orbit 500km alt (r=6878km)

Parameter	Value
Length (a)	0.10 m
Width (b)	0.10 m
Height (c)	0.34 m
Mass (m)	4 kg

Axis	Mol(kg·m ²)
Longitudinal (z)	~0.007
Transverse (x)	~0.041
Transverse (y)	~0.041

Aerodynamic torque Torque: $v = 7.61 \text{ km/s}$; At 500 Km, density $\rho = 1 \times 10^{-11} \text{ kg/m}^3$; cross sec area $= 0.034 \text{ m}^2$, $C_d = 2.2$,

$$F_d = \frac{1}{2} \rho v^2 C_d A = 2 \times 10^{-5} \text{ N}$$

(along the -ve velocity direction)

$$F_{SRP} = -4.5 \times 10^{-6} (1 + r) A$$

$r = \text{Reflection factor } 0 < r < 1$

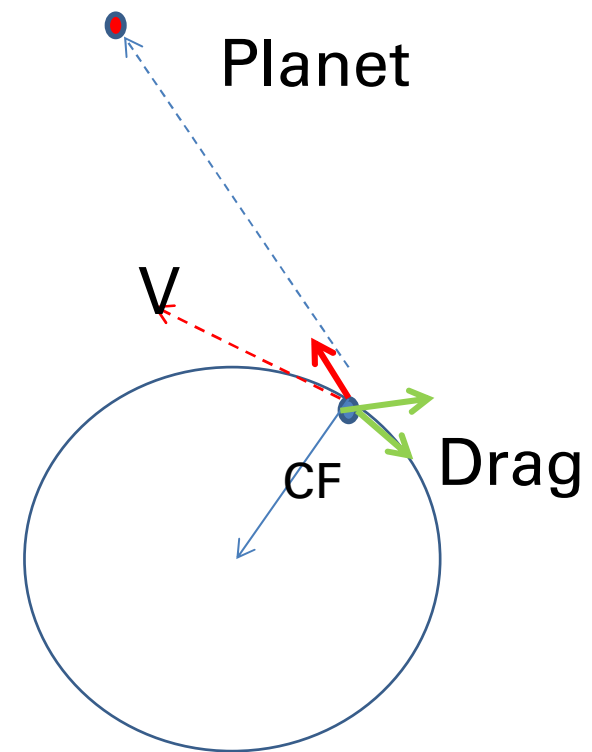
$$\text{For } r=0.4, F = 2.1 \times 10^{-7} \text{ N}$$

Accounting for Perturbations

- Numerical Method
 - Add ALL the accelerations vectorially.

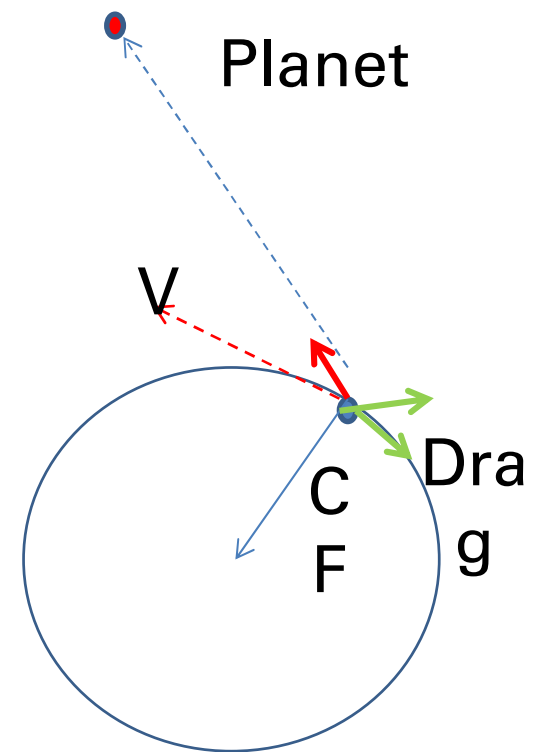
$$\ddot{\vec{a}}_{total} = \ddot{\vec{a}}_{CF} + \ddot{\vec{a}}_{Drag} + \ddot{\vec{a}}_{Planet} + \ddot{\vec{a}}_{SRP} + \dots$$

- Given the Position X, and Velocity V, numerically integrate the acceleration to get new position and velocity.
- Convert the position, velocity to another frame as required.
- Convert the position, velocity to orbital elements as required.



Accounting for Perturbations

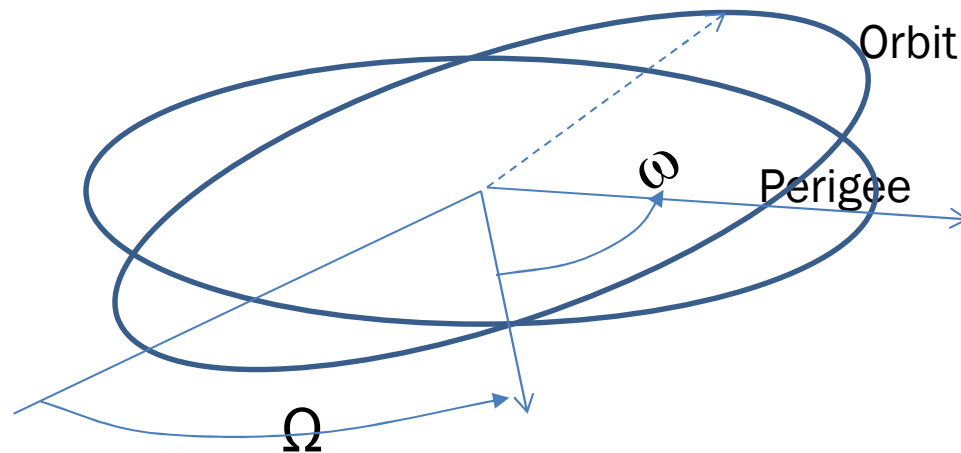
- Analytical Method
 - Using appropriate approximations, derive equations for the time rate of change of orbital elements due to the dominant perturbations.
 $(\dot{a}, \dot{e}, \dot{i}, \dot{\omega}, \dot{\Omega}, \dots)$
 - Advantage: Faster for long term predictions and analysis. They are sufficiently accurate for short term planning of operations.



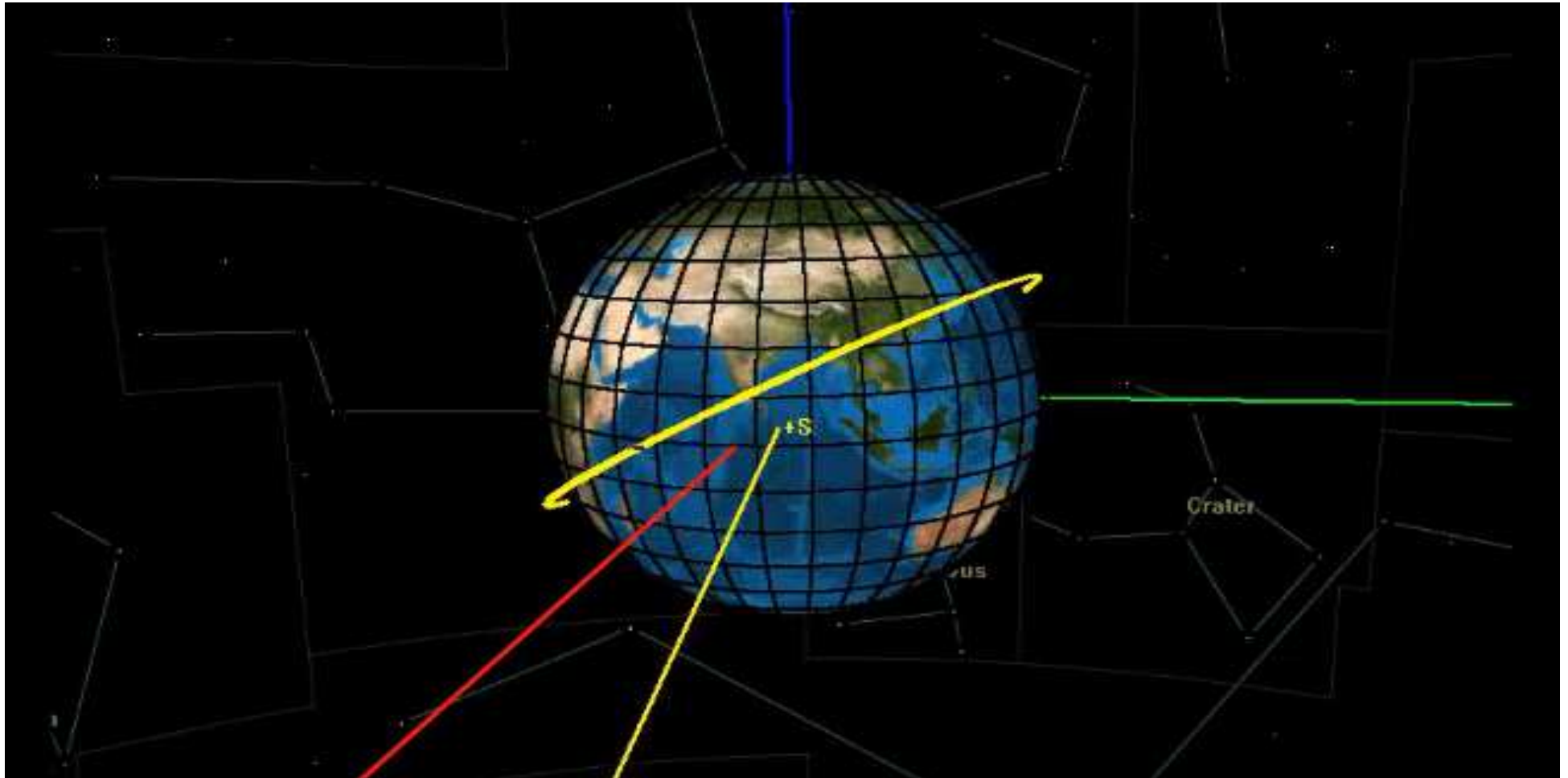
Effect of Orbit Perturbations on Orbital Elements

- Orbit Parameters Change due to
 - Earth Oblateness (non spherical shape, non-uniform mass distribution) Modeled by geo-potential function
 - This causes nodes (Ω) and argument of perigee (ω) to rotate

$$\begin{aligned}\dot{\Omega} &- ve \text{ for } i < 90 \\ \dot{\Omega} &+ ve \text{ for } i > 90\end{aligned}$$

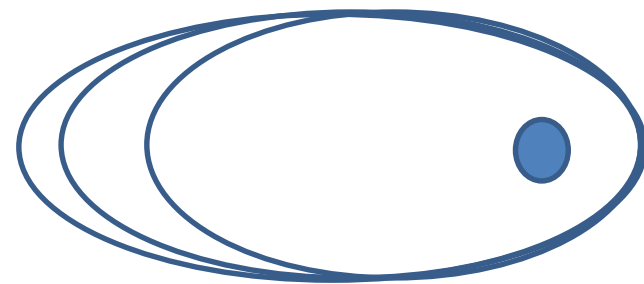


Orbit Regression Animation



Orbit Perturbations (Drag)

- Orbit Parameters Change due to
 - Atmosphere Drag
 - This causes semi-major axis (a) and eccentricity (e) to change
 - The orbit's apogee initially decreases faster and finally becomes circular and then orbit spirals down.



Important Analytical perturbations

- Due to non spherical earth (J2 term only)

$$\begin{aligned}\dot{\Omega}_{J_2} &= -1.5n J_2 (R_E / a)^2 (\cos i) (1 - e^2)^{-2} \\ &\cong -2.064\,74 \times 10^{14} a^{-7/2} (\cos i) (1 - e^2)^{-2} \text{ deg/day}\end{aligned}$$

$$\begin{aligned}\dot{\omega}_{J_2} &= 0.75n J_2 (R_E / a)^2 (4 - 5 \sin^2 i) (1 - e^2)^{-2} \\ &\cong 1.032\,37 \times 10^{14} a^{-7/2} (4 - 5 \sin^2 i) (1 - e^2)^{-2} \text{ deg/day}\end{aligned}$$

Ref: Space Mission Analysis and Design, By Wertz

Important Analytical perturbations

- Due to drag: For near circular orbits.

$$\Delta a_{rev} = -2\pi (C_D A/m) \rho a^2$$

$$\Delta P_{rev} = -6\pi^2 (C_D A/m) \rho a^2 / V$$

$$\Delta V_{rev} = \pi (C_D A/m) \rho a V$$

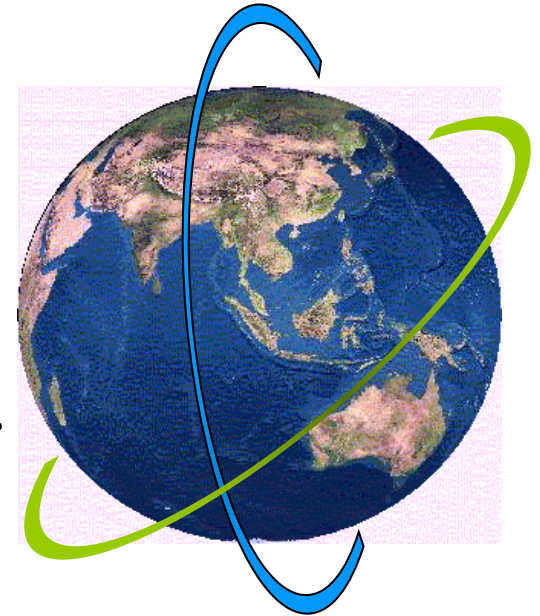
$$\Delta e_{rev} = 0$$

Δa_{rev} , ΔP_{rev} , ΔV_{rev} , Δe_{rev} are the changes/orbit for semi major axis (a), Period (P), Velocity (V) and eccentricity (e).

And $m/(C_D A)$ is the ballistic coefficient

Sun Synchronous Orbit

- Definition: A Sun-synchronous orbit (SSO) is a near-polar orbit where a satellite passes over the same part of the Earth at the same local solar time each day.
- Key Feature: The satellite maintains a consistent relationship (orientation) with the Sun, ensuring near constant lighting conditions for optical imaging.
- Uses:
 - Ideal for Earth observation, weather forecasting, and reconnaissance.
 - The orbit also ensures near constant illumination for the satellite's solar panels, thereby ensuring near constant power generation



Sun Synchronous Orbit-Design

- Mechanism: The orbit is designed to precess (rotate) approximately 1 degree per day (**0.9856deg/day**) to match the Earth's movement around the Sun.
- Inclination and Altitude: The specific inclination and altitude are chosen to achieve the desired precession rate.
- The altitude is then further fine-tuned to achieve repeating ground track pattern
- Examples: Satellites like Landsat, SPOT, and IRS use Sun-synchronous orbits for consistent data collection.

Sun Synchronous Orbit-Design

$$\dot{\Omega} = -\frac{3}{2}J_2\eta \frac{R^2}{a^2(1-e^2)^2} \cos i \text{ (rad/sec)} = 0.9856 \text{ deg/day}$$

Longitude shift (westward) between successive orbits $\Delta\lambda = (w_e - \dot{\Omega})T_N$

If the ground tracks repeat after M orbits in N days, then $(\Delta\lambda)M = 2\pi N$

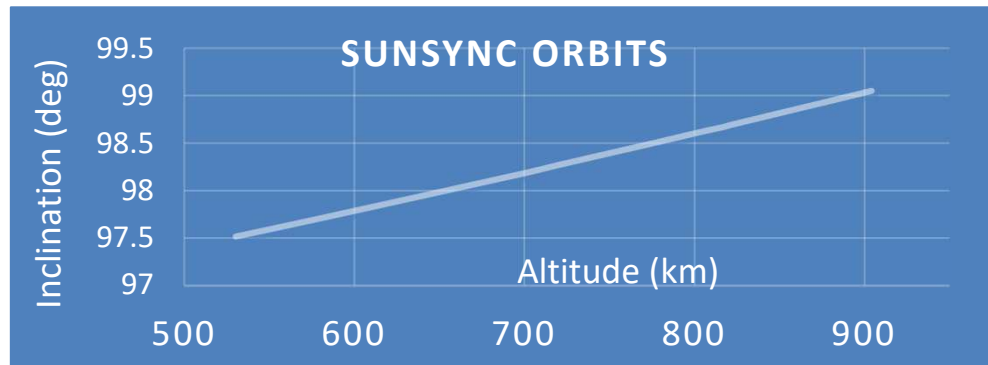
At ~ 530 km altitude, a suitable SSO is,

$h = 530.25\text{km}$, $i = 97.517 \text{ deg}$.

The ground trace repeats in 10 days after 151 orbits.

Sun Synchronous Orbits

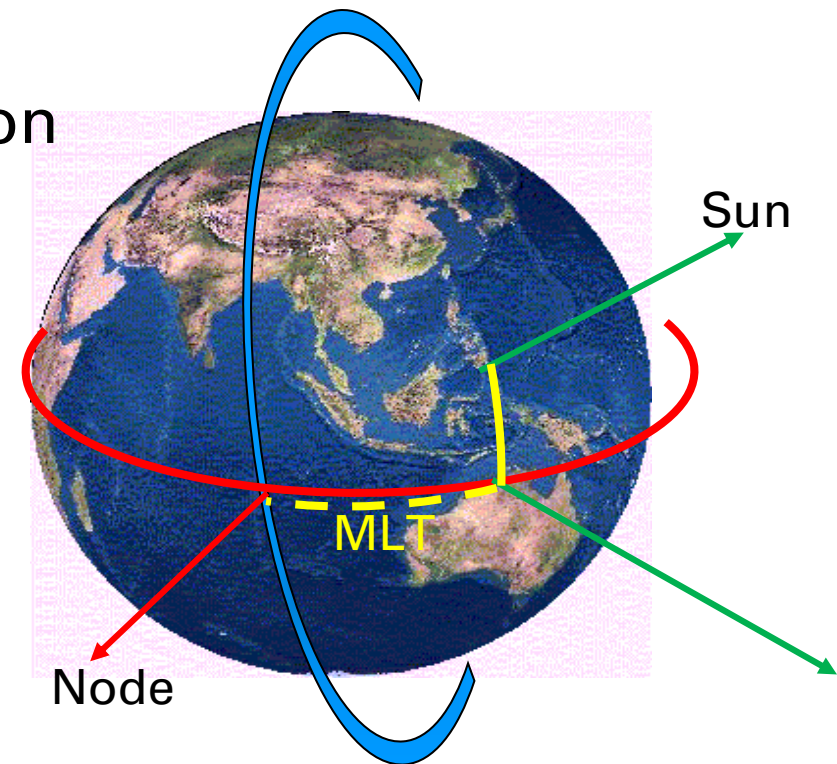
h (km)	i (deg)	Repetition		Remarks
		orbits	days	
530.25	97.517	151	10	15+(1/10)
561	97.635	15	1	15
592	97.756	149	10	15-(1/10)
699	98.18	233	16	Landsat, 14+(9/16)
719.9	98.27	29	2	14+(1/2)
775	98.497	43	3	14+(1/3)
802.9	98.615	57	4	14+(1/4)
817	98.67	341	24	IRS-1C, 14+(5/24)
822.4	98.697	369	26	Spot, 14+(5/26)
888	98.98	14	1	14
904.12	99.051	307	22	IRS-1A, 14-(1/22)



Mean Local Time (MLT) for SSO

MLT in deg is,
= 0, the node local time is 12hrs noon
= 15, the node local time is 11 am
= 30, the node local time is 10 am

Chosen as per the imaging
requirements



Case Study: 3U CubeSat

Orbit 500km alt (r=6878km)

Parameter	Value
Length (a)	0.10 m
Width (b)	0.10 m
Height (c)	0.34 m
Mass (m)	4 kg

Aerodynamic Force:

$v = 7.61 \text{ km/s}$; Alt 500 Km,
Density $\rho = 1 \times 10^{-11} \text{ kg/m}^3$;
Cross sec area $= 0.034 \text{ m}^2$,
 $C_d = 2.2$,

$$F_d = \frac{1}{2} \rho v^2 C_d A = 2 \times 10^{-5} \text{ N}$$

(along the -ve velocity direction)

$\Delta a \text{ per orbit} \cong 50 \text{ m OR}$
 700 m/day

For low altitudes, decay is dominant. The density assumed is on the higher side. If $\rho \sim 1 \times 10^{-12}$ then decay will be around 70m/day

ORBIT MANEUVERS

Why Orbit Maneuver?

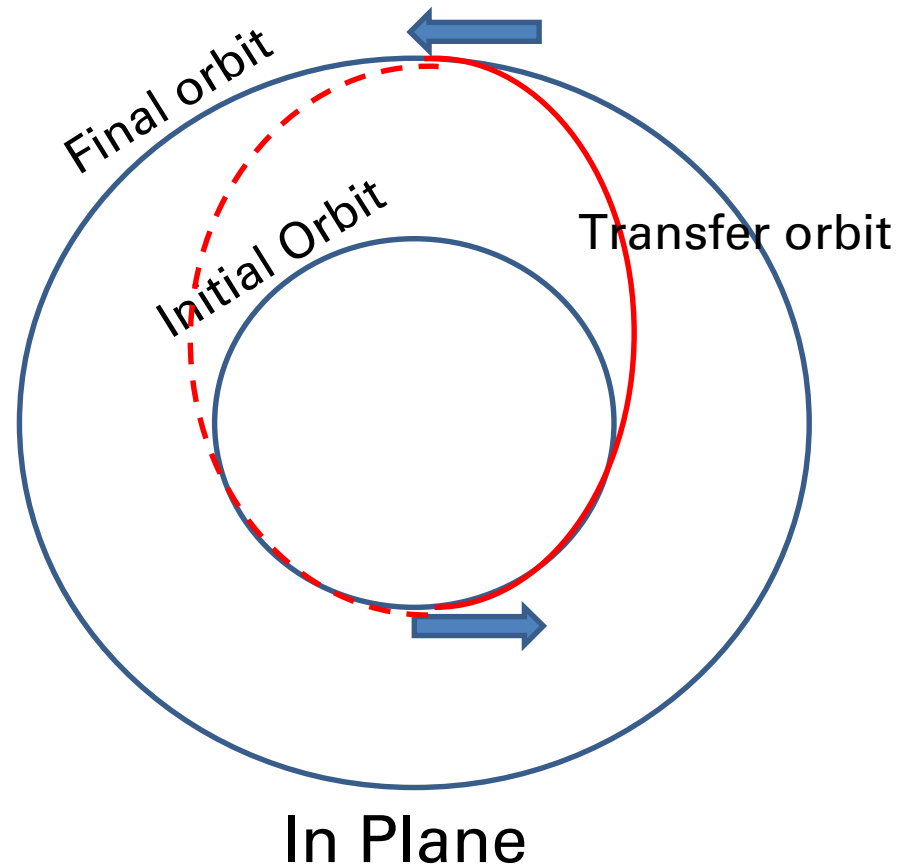
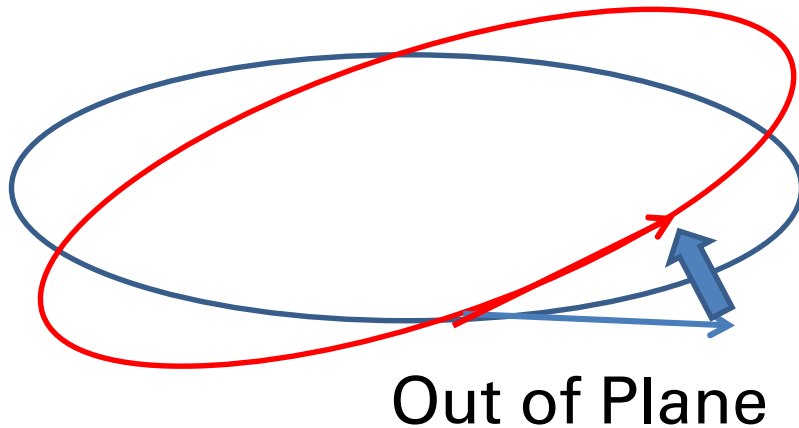
- **Orbit Maneuver (small or large) is required for,**
 - Removing the launcher injection errors to realize the nominal orbit.
 - Placing the satellite in an orbit different from the launcher orbit.
 - Maintaining a nominal orbit against perturbations
 - For achieving different mission goals during the mission life

Orbit Maneuver Types

Type	Description
Transfer	To go from an initial to final orbit (e.g. LEO to GTO to GEO)
Trim	To make small corrections (e.g. Launch error corrections)
Alter Ground Track	To change the ground track pattern to pass over a desired target
Compensate	To compensate for perturbation (i.e. orbit maintenance)
Acquire in phase	To alter and reacquire the orbit to achieve spacing between satellites in a constellation

Orbit Transfer

- Orbit Transfer
 - In Plane
 - Out of Plane

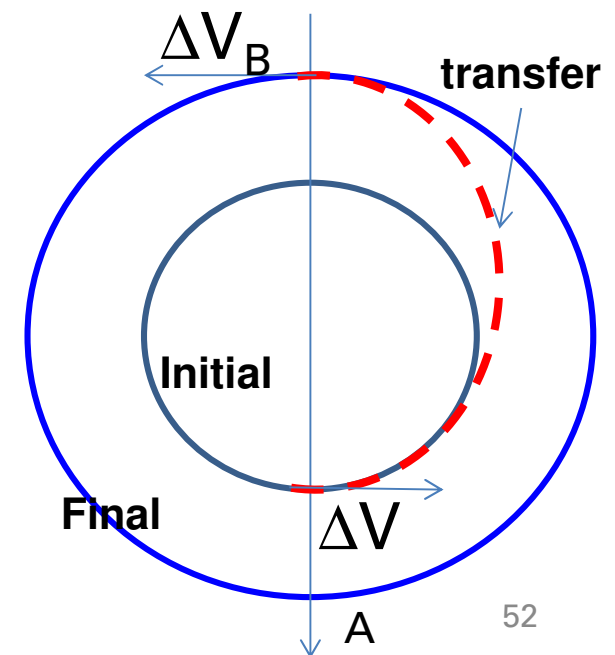


In-Plane Transfer ΔV Calculations

- The velocity increment is the vector difference between the velocity vectors of initial and final orbits. $\overline{\Delta V} = \overline{V}_{final} - \overline{V}_{initial}$
- The transfer orbit that is tangential to initial as well as final orbit is known as 'Hohmann Transfer'
- Hohmann Transfer is fuel efficient
- ΔV is realized by firing thrusters, with force F
- Now, $F = \frac{d}{dt}(MV)$
- Using the propellant property I_{sp} (Specific Impulse), the rocket equation gives ΔV

$$\Delta V = g_0 I_{sp} \ln \left(\frac{WT}{WT - WF} \right) \text{ where}$$

WT total mass, WF fuel mass



Typical in-plane transfer from LEO to GEO

- $\Delta V1 = V_{P,TO} - V_{LEO}$
- $\Delta V2 = V_{GSO} - V_{A,TO}$

Example:

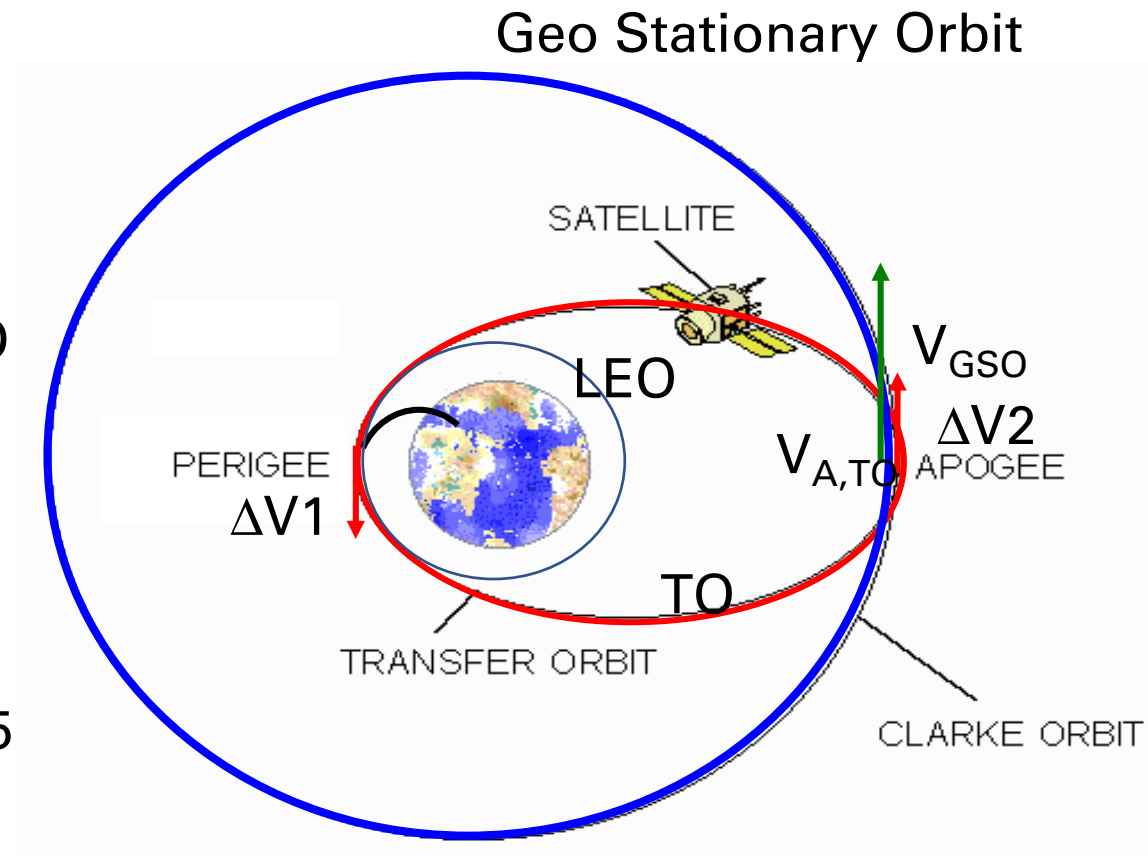
Transfer from 500 km LEO to GSO

LEO: $r = 6878$ kms; $V_{LEO} = 7.612$ km/s

TO: $r_p = 6878$, $r_a = r_{gso} = 42164$ kms

GSO: $r_{gso} = 42164$ kms; $V_{gso} = 3.075$ km/s

Calculate the perigee, apogee velocities and $\Delta V1$ and $\Delta V2$



[Animation](#)

Orbital Plane Change Maneuver

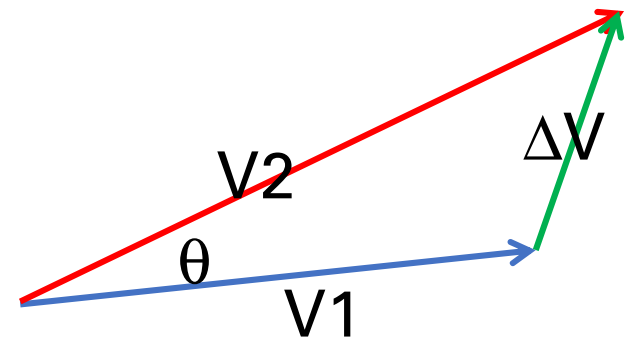
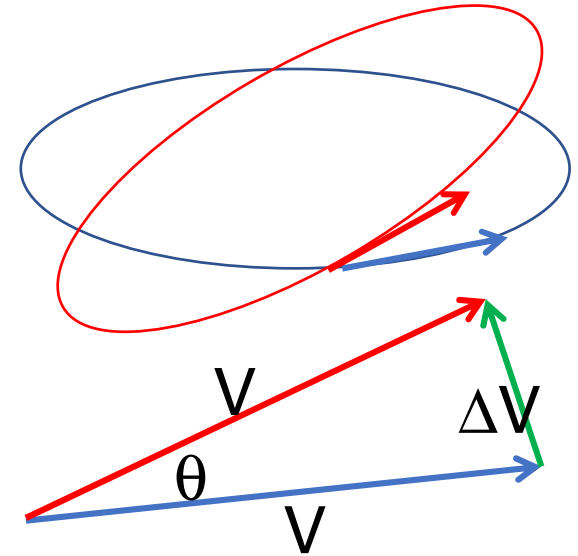
- Simple Plane Change

- The orbital radius before and after is same

- $\Delta V = 2V \sin \frac{\theta}{2}$

- Combined Plane Change

- $\Delta V = \sqrt{(v1)^2 + (v2)^2 - 2(v1)(v2)\cos\theta}$



Small Maneuver Approximations

- For circular orbits, $v^2 = \frac{\mu}{r}$;
- For in-plane maneuver, $\Delta v = -\frac{v}{2r}\Delta r$
- For out-of-plane maneuver, $\Delta v = v\Delta\theta$
- For LEO, at 500km, $r=6878$ km, $v=7.61$ km/s
- $\Delta V = 0.55$ m/s per km change in r (in-plane)
- $\Delta V=0.55$ m/s causes a change of 0.004 deg in inclination

*Thumb Rule: For LEO, ΔV of **1 m/s** causes **2km** change in orbit radius*

TWO LINE ELEMENTS (TLE)

Two Line Elements (TLE)

- Widely used to exchange orbital information between different agencies
- Well developed analytical propagators available in the public domain.
- Accuracy sufficient for mission planning and satellite tracking
- Orbital information is regularly updated and maintained by the international body

TLE Description

Year

Day of yr

Time

```
1 37389U 11015C 12117.68129472 .00000367 00000-0 17958-3 0 1349
2 37389 098.7351 193.4675 0013066 304.6376 055.3557 14.23886679 52994
```

i

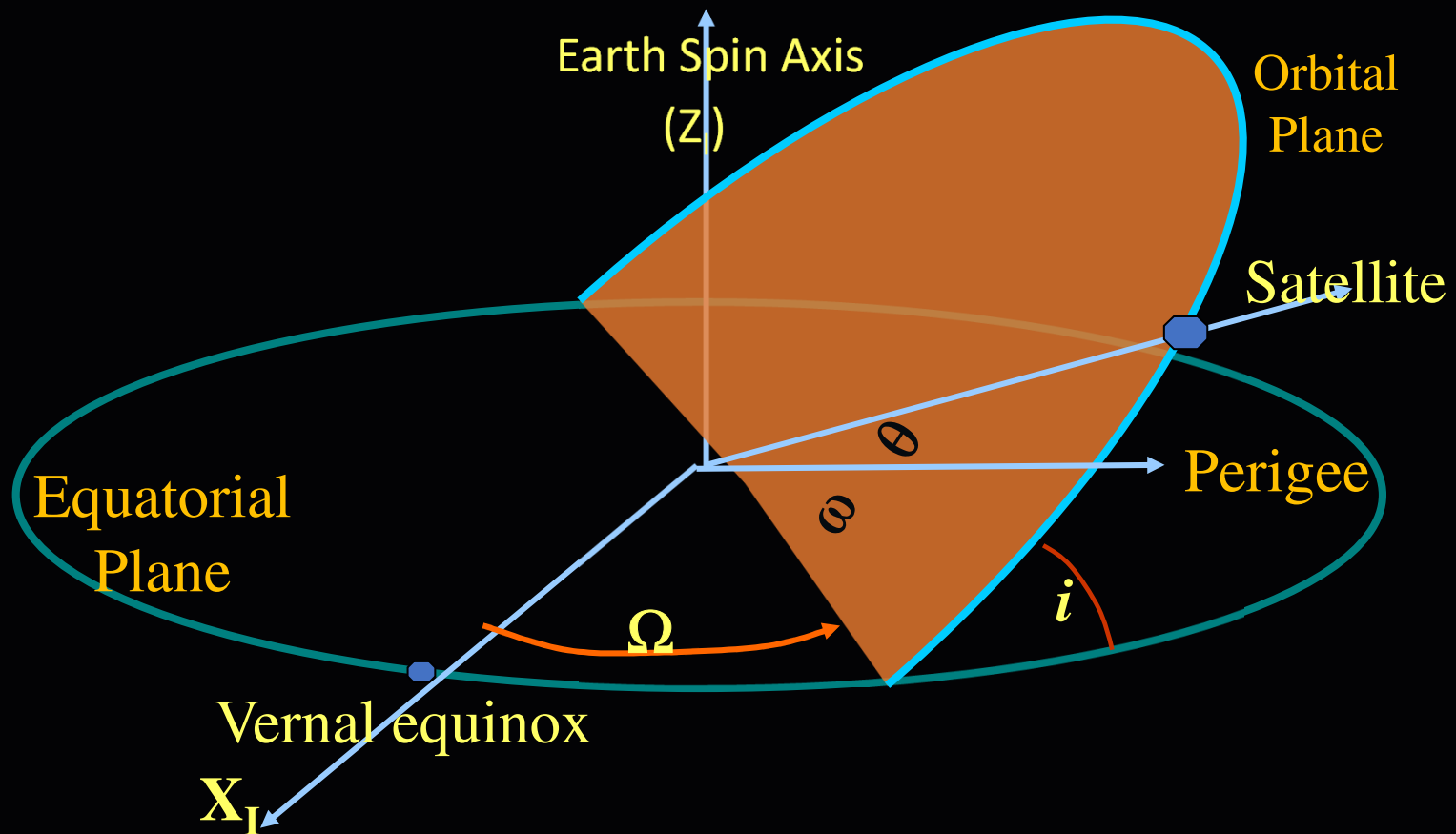
Ω

ω

M

Rev/day

RevNo



Typical TLEs

XSAT (LEO, SunSync Orbit)

```
1 37389U 11015C 25179.93255193 .00000057 00000-0 41762-4 0 9995
2 37389 98.3119 169.6391 0012995 303.5976 56.3963 14.25046573 737362
```

INSAT 3D (GEO Orbit)

```
1 39216U 13038B 18228.86865342 -.00000179 00000-0 00000+0 0 9996
2 39216 0.0919 279.7967 0001198 245.8146 194.3778 1.00271331 18611
```

NAVSTAR 70 (MEO Orbit)

```
1 39741U 14026A 18229.11404031 -.00000060 00000-0 00000+0 0 9999
2 39741 55.6867 83.3297 0011977 293.3274 66.6166 2.00562820 31134
```

MOLNIYA 1-49(HEO Orbit)

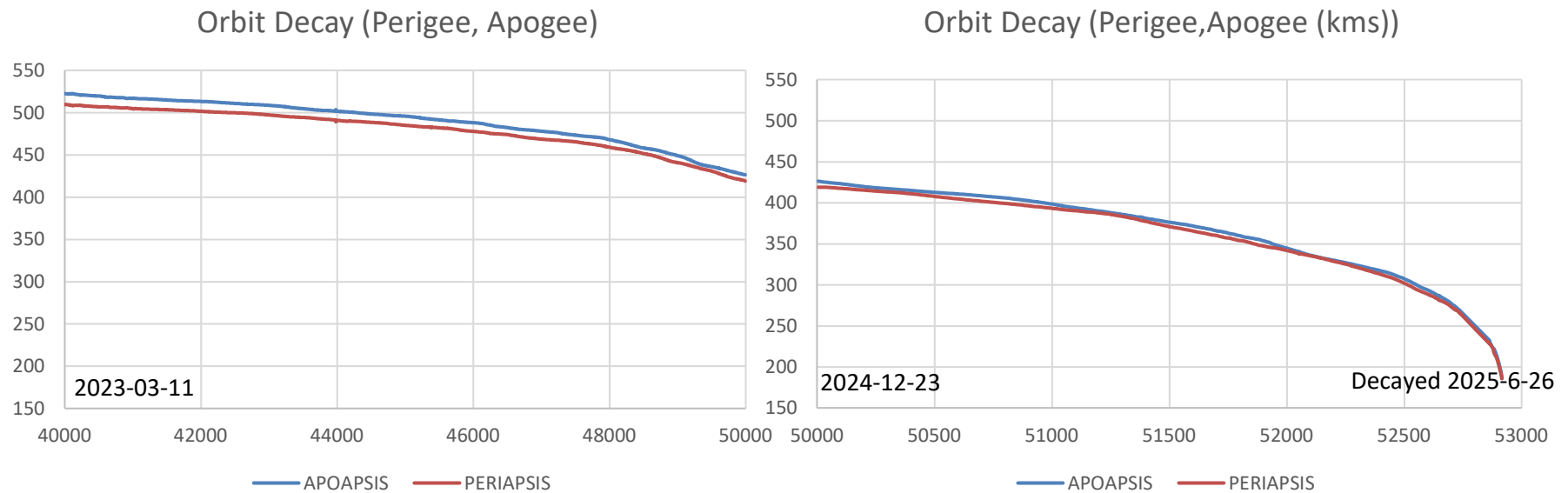
```
1 12156U 81009A 18231.08835293 .00000701 00000-0 26946-2 0 9993
2 12156 63.5722 15.1450 7214153 247.3961 25.8002 2.05696146192753
```

Get TLE from, <https://www.space-track.org/>

TLE Based Analysis

TeLEOS-1 Orbit Decay

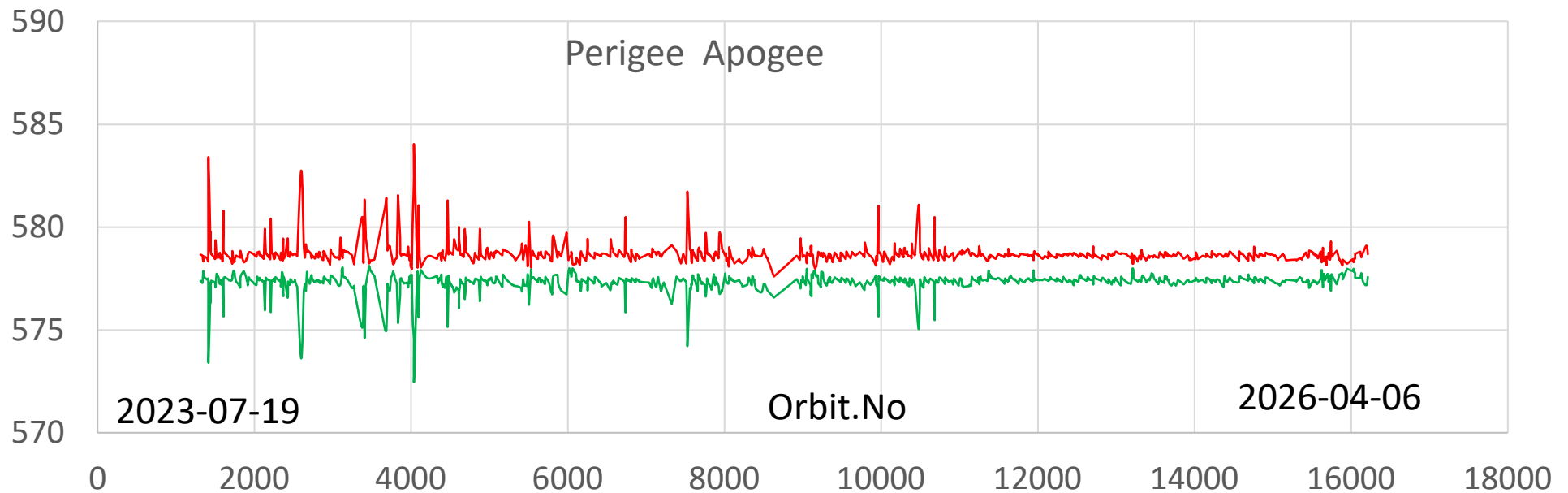
- Re-entered on 2025-06-26



TLE Based Analysis

TeLEOS-2 Orbit History

- Launched in Apr 2023
- Orbit Maintained to keep the ground track pattern within bounds



Orbit Determination

Orbit Determination (OD) is a process of estimating the orbital parameters based on observations of satellite over a span of time.

OD is needed because

1. The launch errors result in an orbit different from the desired orbit.
2. The perturbations keep changing the orbit and thus slowly drift from the nominal orbit.
3. Accurate orbit information is needed to plan the mission operations for a few days ahead
4. Accurate orbit information is needed to process the satellite payload data analyze the contents.
5. Accurate orbit information is needed to plan the DV correction for orbit maintenance

Orbit Determination Principle (by Ground Tracking)

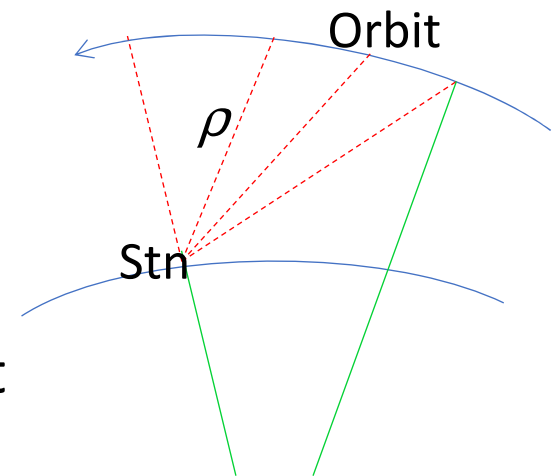
The satellite is tracked from a ground station and the observations range (ρ), range rate ($\dot{\rho}$), Azimuth(ζ) and Elevation(σ) are collected for a span of time. Let Z be the set of observations. Then if E is the orbital element ,

$$Z_t = f(E) = f(E_0, t) \quad \text{where } E_0 \text{ is the Element set at epoch } t_0$$

Starting with an initial guess for E_0 ,

$$\Delta Z = \frac{\partial Z}{\partial E_0} \Delta E_0$$

Using a Batch Least Squares estimator, the element set E_0 is obtained from the measurement set



Orbit Determination Principle (onboard GPS Rx)

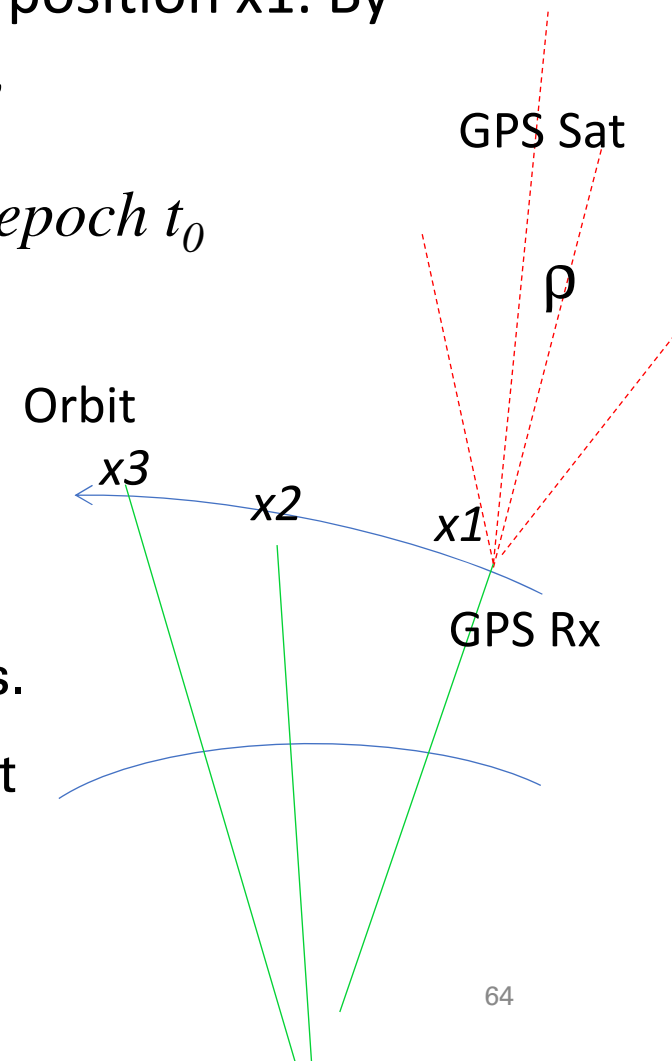
The satellite's onboard GPS receivers collect measurements (ρ) from GPS constellation and estimates the instantaneous position x_1 . By using $x_1, x_2 \dots x_t$ or ρ as measurement set Z , we get,

$$Z_t = f(E) = f(E_0, t) \text{ where } E_0 \text{ is the Element set at epoch } t_0$$

Starting with an initial guess for E_0 ,

$$\Delta Z = \frac{\partial Z}{\partial E_0} \Delta E_0$$

- x_1, x_2, x_3 are used onboard for realtime operations.
- Using a Batch Least Squares estimator, the element set E_0 is obtained from the measurement set



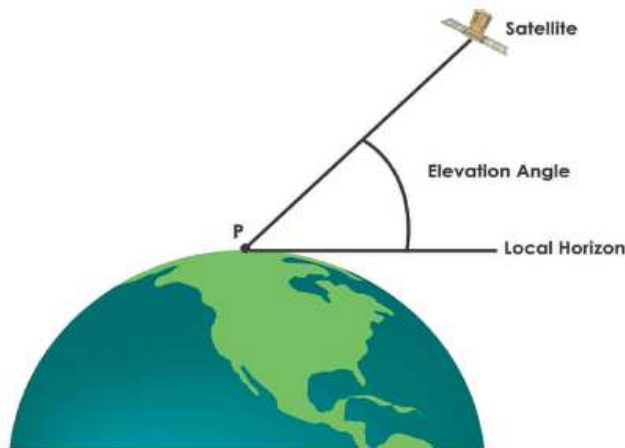
How orbital elements are used for day-to-day mission operations....

ORBITAL EVENTS

Orbital Events

- Visibility
- Eclipse
- Nodes, Apsides crossing
- Ground Tracks

Coverage from a satellite



When the satellite is above the horizon of an observer, the satellite is said to be visible to the observer

The cone defines the area of coverage from an altitude.

The higher the altitude, the larger shall be the coverage area



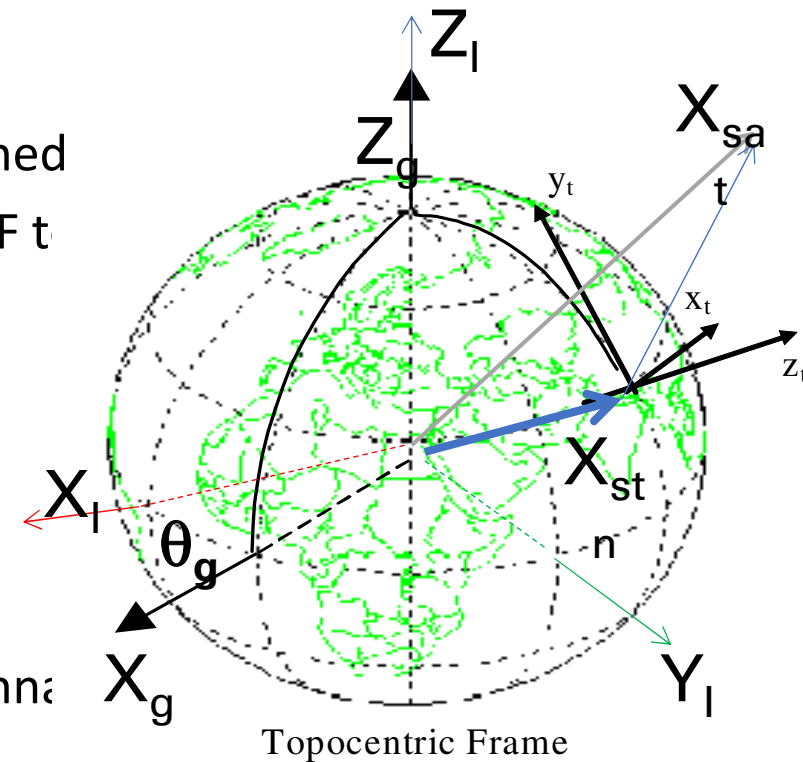
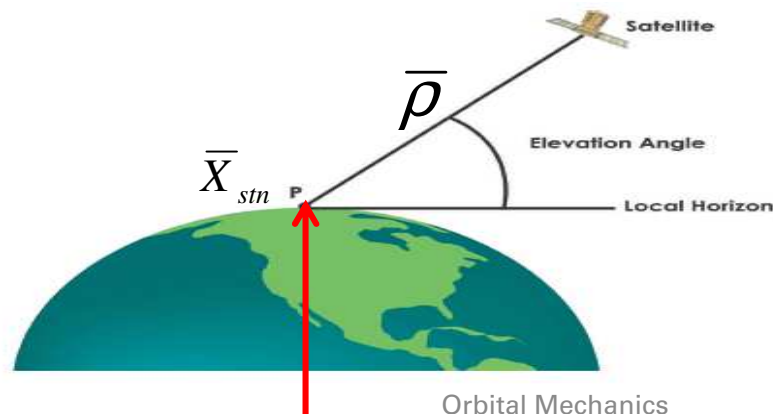
Visibility Calculation

- X_{sat} , Satellite position in inertial frame
- X_{stn} , Station position in inertial frame (obtained by transforming the station vector from ECEF to ECI frame)

$$\bar{\rho} = \bar{X}_{sat} - \bar{X}_{stn}$$

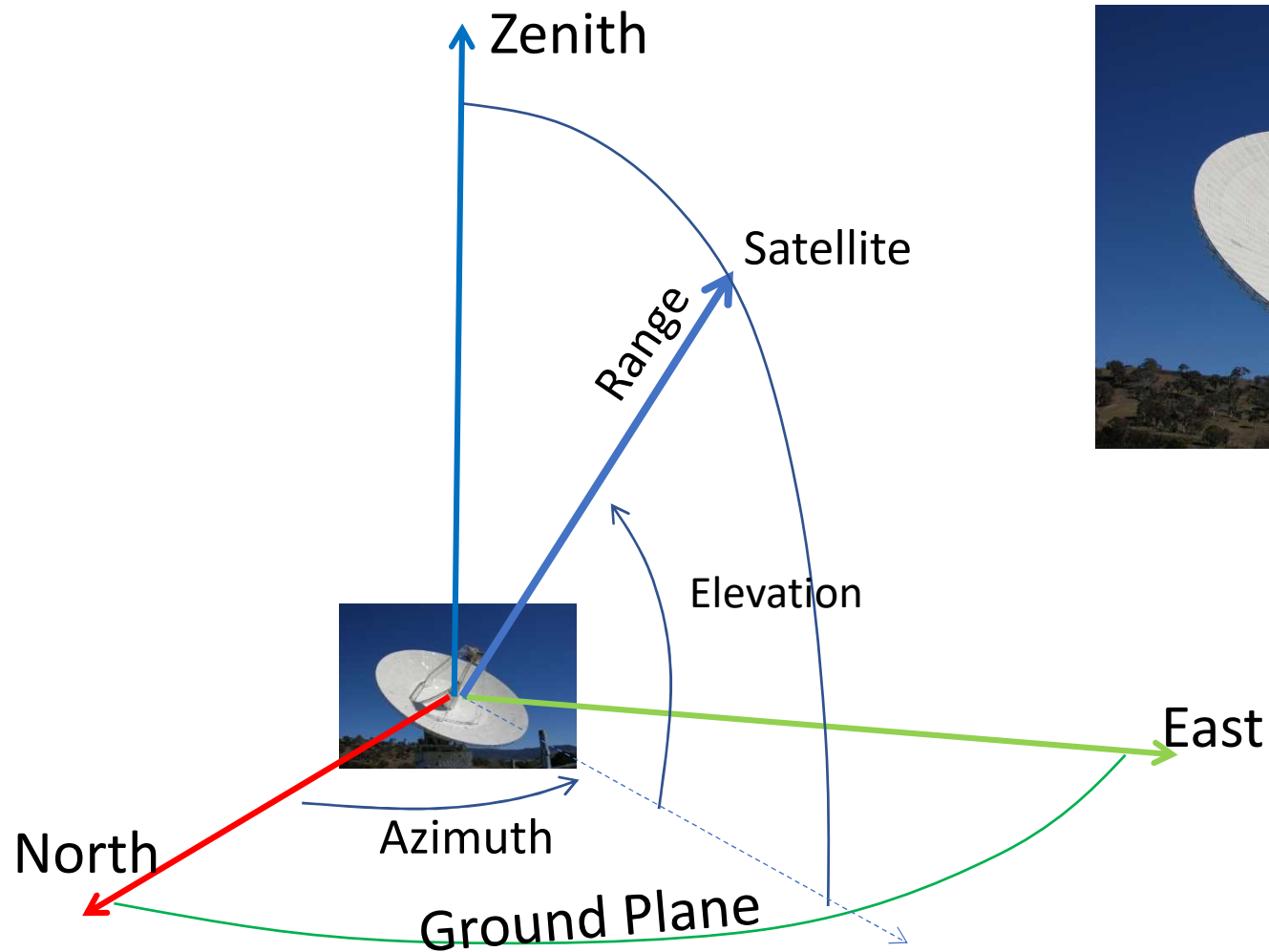
If $\bar{\rho} \cdot \bar{X}_{stn} > 0$, Satellite Visible

- Then transform $\bar{\rho}$ to topocentric frame and compute range, azimuth, elevation for antenna tracking



$[XYZ]_I$ ECI frame
 $[XYZ]_g$ ECEF Frame
 $[XYZ]_t$ Topocentric Frame

Topocentric Frame



Orbital Eclipse

- When a satellite moves around Earth, at times it gets into the shadow of earth. During these times, the satellite operations wholly rely on its onboard stored energies like battery.
- As part of mission planning and operations, it is essential to predict such events of satellite going through the shadow to operate the satellite and payload safely.
- The orbital eclipse event is governed by the positions of satellite, Sun and Earth.

Orbital Eclipse Calculation



If the angle between satellite-earth and satellite-sun unit vectors (α) is greater than α_0 then satellite is under sun-light; otherwise it is under eclipse. That is,

$$\left(-\frac{\bar{X}_{sat,I}}{|\bar{X}_{sat,I}|} \right) \cdot \left(\frac{\bar{X}_{sun,I}}{|\bar{X}_{sun,I}|} \right) < \cos(\alpha_0)$$

The satellite position is obtained by orbit propagation and sun vector is obtained from sun ephemerides

Maximum Eclipse in an Orbit

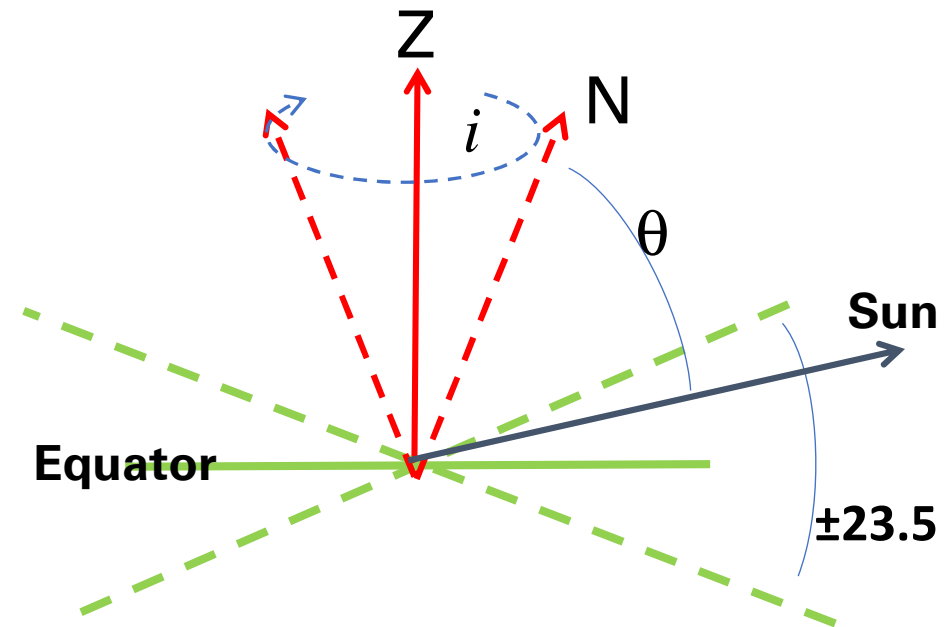
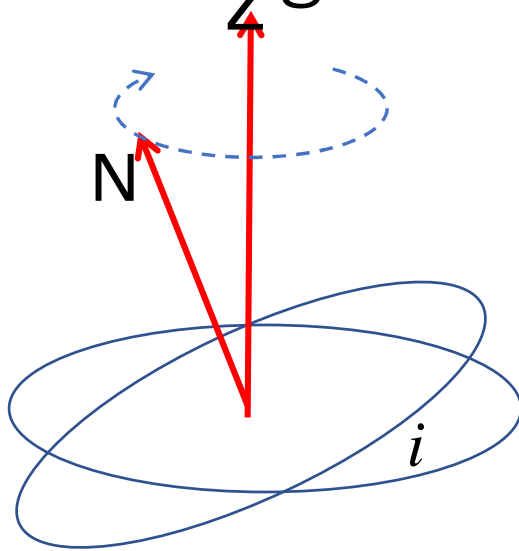
Altitude (kms)	Period (minutes)	Eclipse (minutes)	Eclipse orbit	(% of	α_0 (deg)
550	95.647	35.61	37.23		67.0
1000	105.11	34.93	33.23		59.8
10000	347.6	44.26	12.73		22.9
35786	1436	69.41	4.83		8.7



Note:

1. The maximum eclipse occurs when Sun is in the orbital plane.
2. When the separation between orbit plane and sun $> \alpha_0$, there will be no eclipse.

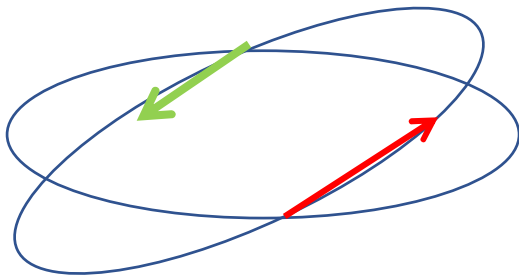
Sun Angle Variations



Due to orbit regression and sun apparent motion with season, Sun angle (θ) w.r.t orbit normal(N) changes every day.

Minm : $90-(i+23.5)$; Max: $90+ (i+23.5)$;

Node, Apogee, Perigee crossing



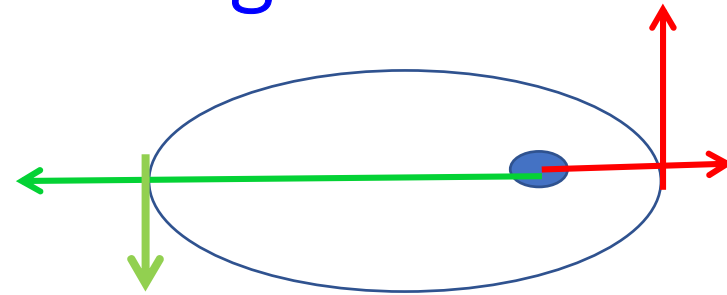
Conditions:

$Z = 0, \dot{Z} = +ve$ for **Asc.node**

$Z = 0, \dot{Z} = -ve$ for **Dsc.node**

Appls:

Orbit.no counting, Inclination change maneuvers



Conditions:

$r \cdot V = 0, |V| = \text{Max}$ for **Perigee**

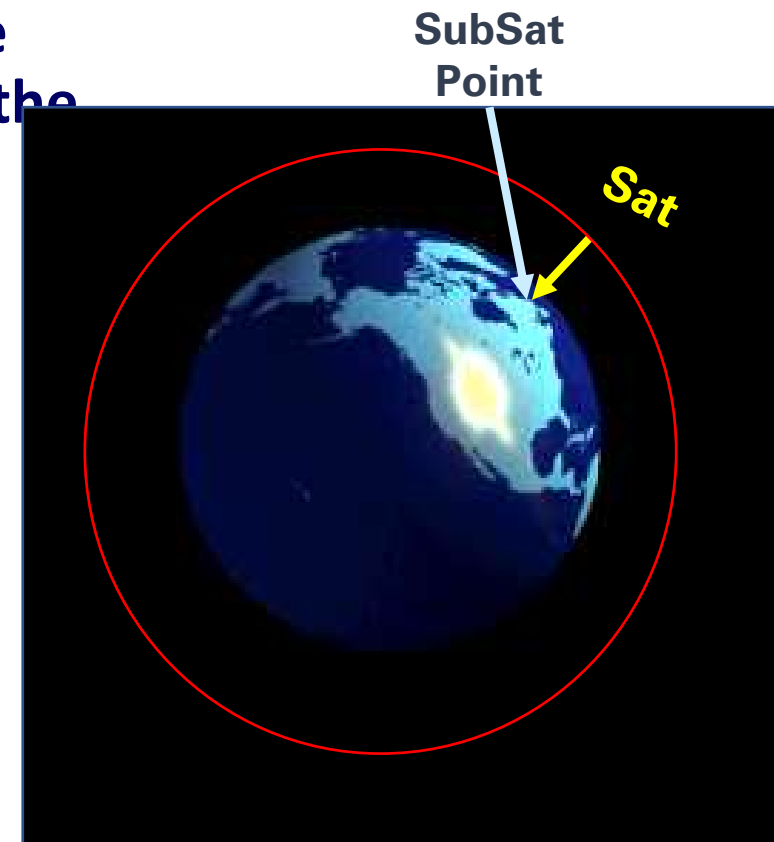
$r \cdot V = 0, |V| = \text{Min}$ for **Apogee**

Appls:

Hohmann transfer maneuvers

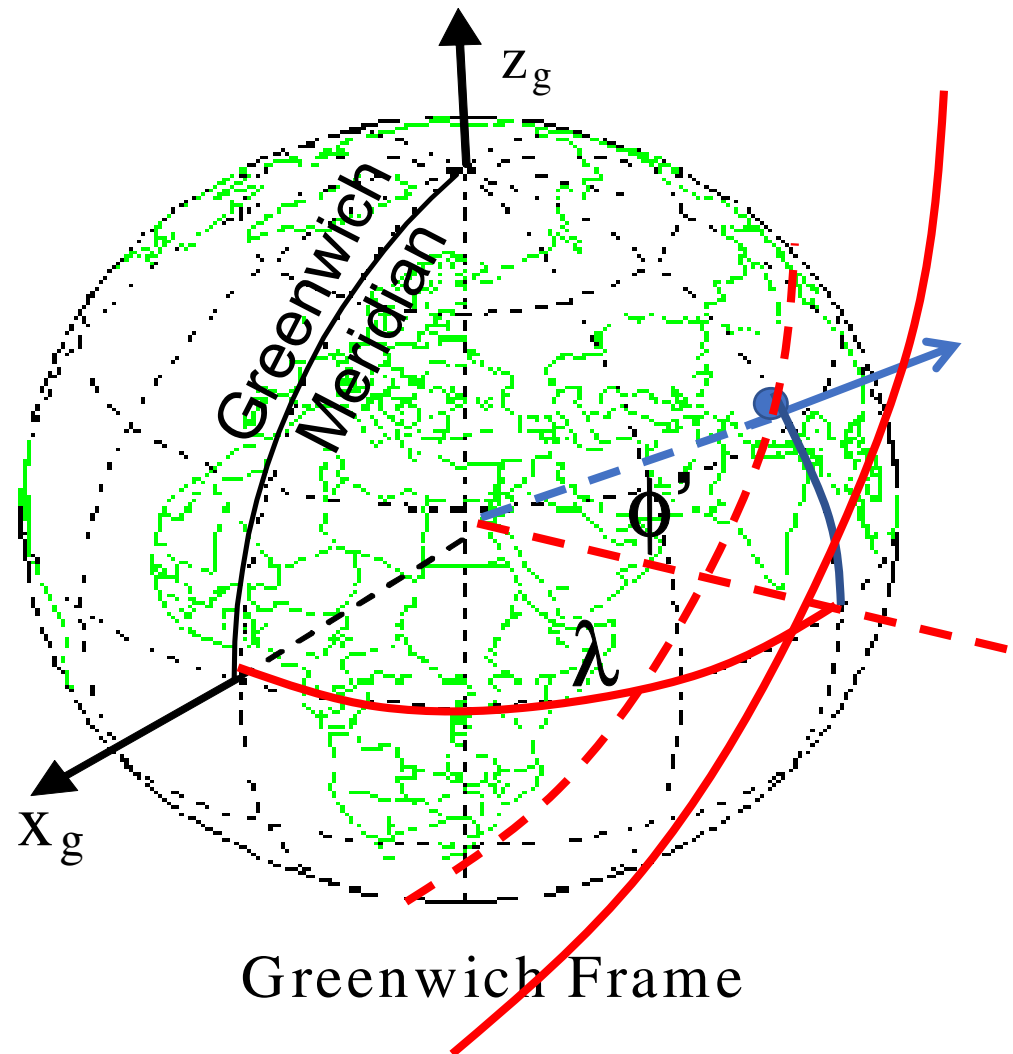
Ground Track

- **Projection** of satellite position **onto** the surface of **earth** is the **sub-satellite point**.
- **Locus** of sub-satellite point is the **Ground Track**
- Ground Tracks are useful to understand the motion of satellites over different parts of the globe.
 - Useful for planning areas for imaging
 - Communications coverage



Geocentric Coordinates

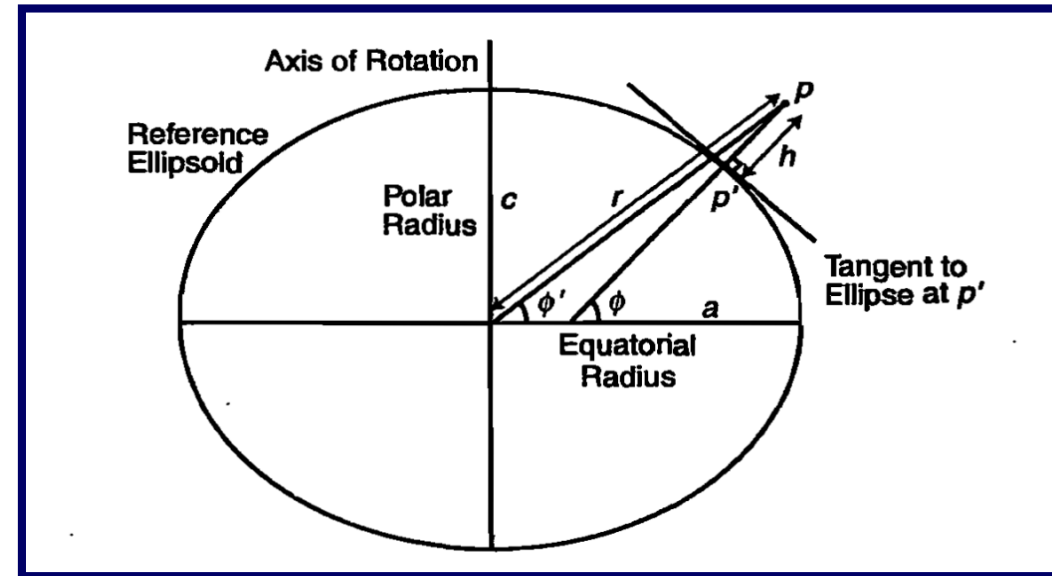
- λ the longitude is measured from Greenwich meridian
- ϕ' the geo centric latitude



Definition of Latitude

- ϕ' the geocentric latitude
- ϕ the geodetic latitude

$$f = \frac{a - b}{a}$$



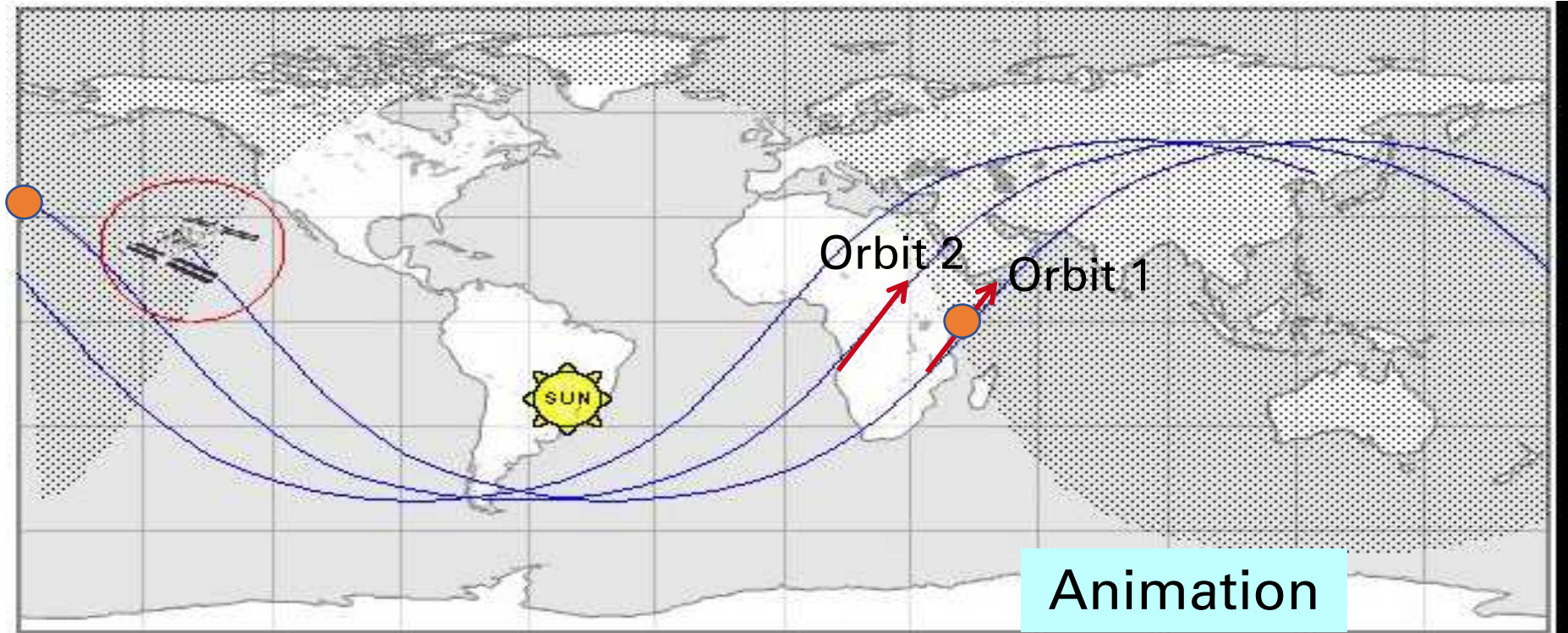
Ellipsoid reference	Semi-major axis (a)	Semi-minor axis (b)	Inverse flattening (1/f)
WGS 84	6 378 137.0 m	≈ 6 356 752.314 245 m	298.257 223 563

Lat,Long to Cartesian Coordinates

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_p = \begin{bmatrix} (N_\phi + h) \cos \phi \cos \lambda \\ (N_\phi + h) \cos \phi \sin \lambda \\ ((1 - e^2)N_\phi + h) \sin \phi \end{bmatrix} \quad \begin{aligned} e &\equiv \sqrt{a^2 - b^2} / a = \sqrt{2f - f^2} \\ N_\phi &\equiv a / \sqrt{1 - e^2 \sin^2 \phi} \end{aligned}$$

The $[X, Y, Z]_p$ components are in Greenwich Meridian Frame, also known as Earth Centered Earth Fixed (ECEF) frame

Ground Track of LEO circular orbits



For a LEO orbit, the orbital period is ~ 100 minutes.

For inclination < 90 deg, the satellite foot print moves from west to east at the equator.

During one orbit (~ 100 min), the earth rotates by 25 deg. So, the second orbit is to the west of orbit 1

LEO ORBIT CONSTELLATION

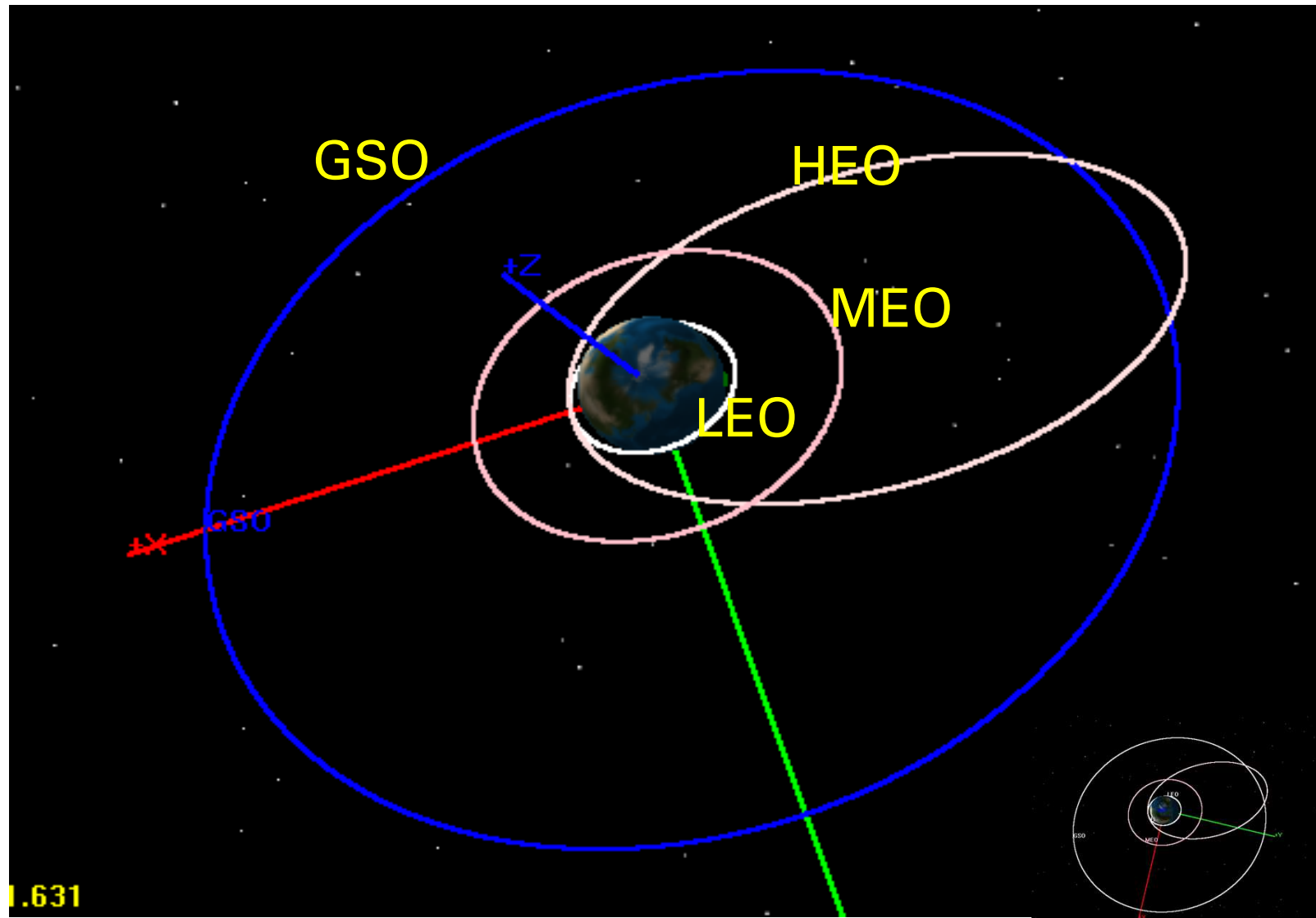
Constellation Design

- Constellations - Definition and Types
- Key Orbital Equations
 - Longitude shift per orbit , nodal period, Orbit Regression
- Constellation design inputs
 - Altitude, minimum elevation
 - Latitudes of coverage
- Constellation Design –Street of coverage, Snap-Shot Coverage, Phasing in adjacent planes.
- Walker Constellations
- Constellation Establishment and Maintenance

Constellation Definition

- A constellation consists of one or more orbital planes with each plane having one or more satellites uniformly distributed in the plane.
- Types of Constellation
 - LEO Constellation (Low Earth Orbit)
 - MEO Constellation (Medium Earth Orbit)
 - GSO Constellation (Geo Synchronous Orbit)
 - HEO Constellation (Highly Elliptical Orbits)

Types of Constellation



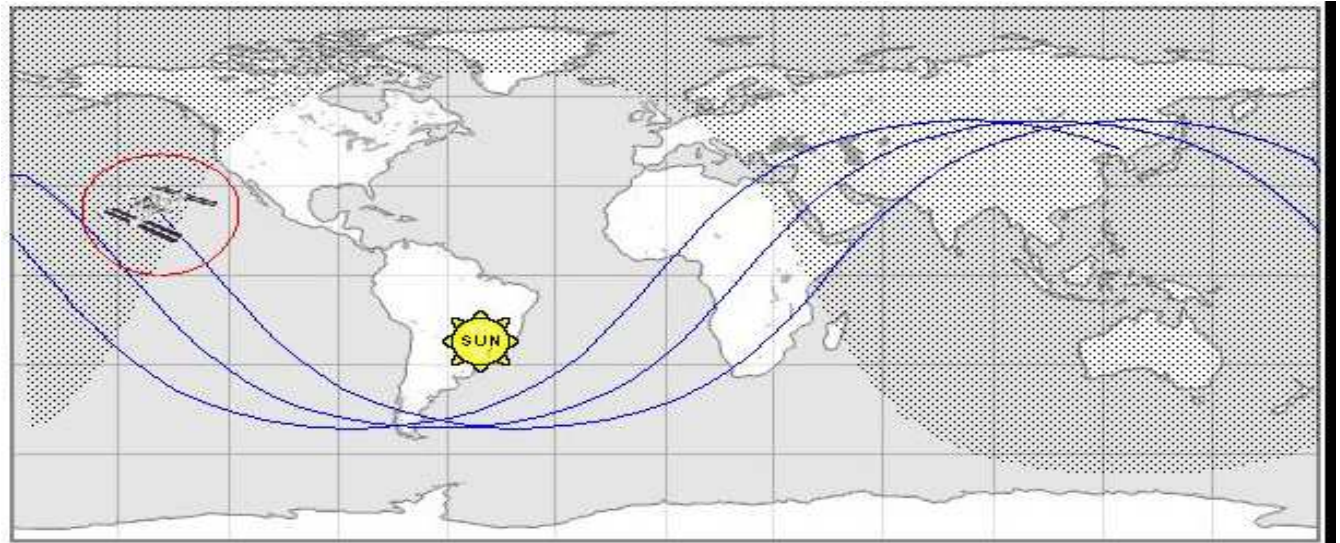
LEO Constellation for communications

- Orbits are circular in general
- They move in inclined orbital planes (low, medium or high inclined planes).
- Planes are uniformly distributed along the equator
- Satellites are equally spaced in the orbital plane.

Orbit Equations for constellation

- Key Equations

- Longitude shift (westward) between successive orbits $\Delta\lambda = (w_e - \dot{\Omega})T_N$
- Nodal Period $T_N = \frac{2\pi}{\eta + \dot{\omega}}$
- $\eta, \dot{\omega}, \dot{\Omega}$ are the mean motion, perigee rotation rate, orbit regression
- The equations consider the effect of J2 to start with



Relevant Perturbation Equations

$$\eta = \eta_0 \left[1 - \left(\frac{3}{2} J_2 \frac{R^2}{a^2 (1 - e^2)^{3/2}} \right) \left(\frac{3}{2} \sin^2 i - 1 \right) \right]$$

$$\dot{\omega} = \frac{3}{2} J_2 \eta \frac{R^2}{a^2 (1 - e^2)^2} \left(2 - \frac{5}{2} \sin^2 i \right)$$

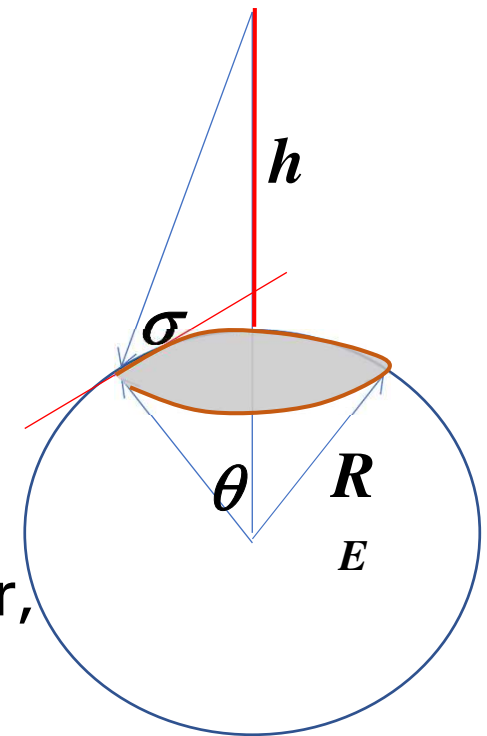
$$\dot{\Omega} = -\frac{3}{2} J_2 \eta \frac{R^2}{a^2 (1 - e^2)^2} \cos i$$

Orbit Design Inputs

- Extent of latitude to be covered (upto poles, near equatorial...)
- Altitude

Latitude Span  Inclination

Altitude  Coverage angle,
Coverage corridor,
No.of planes..



Single Satellite Coverage

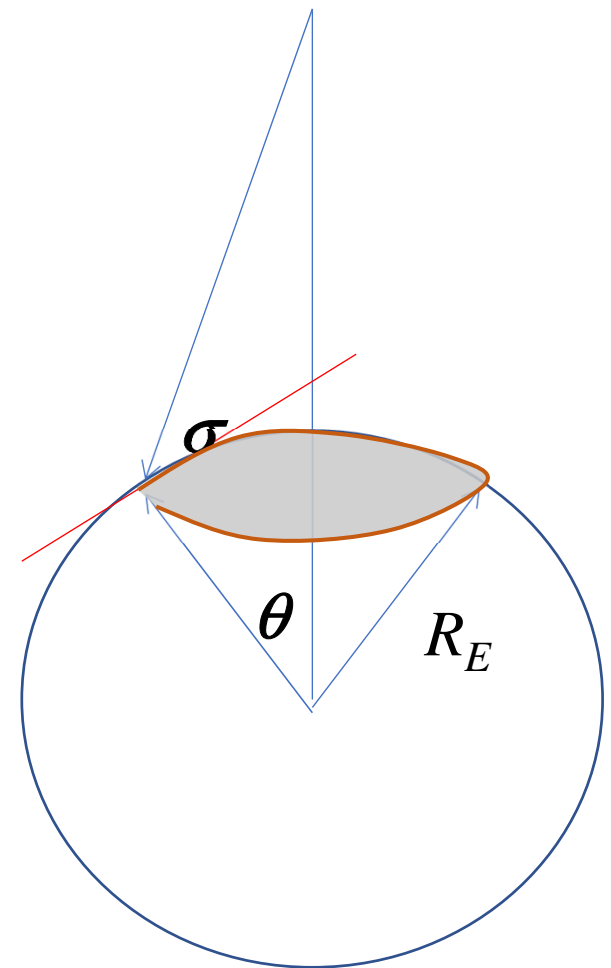
$$\theta = \cos^{-1} \left(\frac{\cos \sigma}{1 + \frac{h}{R_E}} \right) - \sigma$$

The coverage area S is given by,

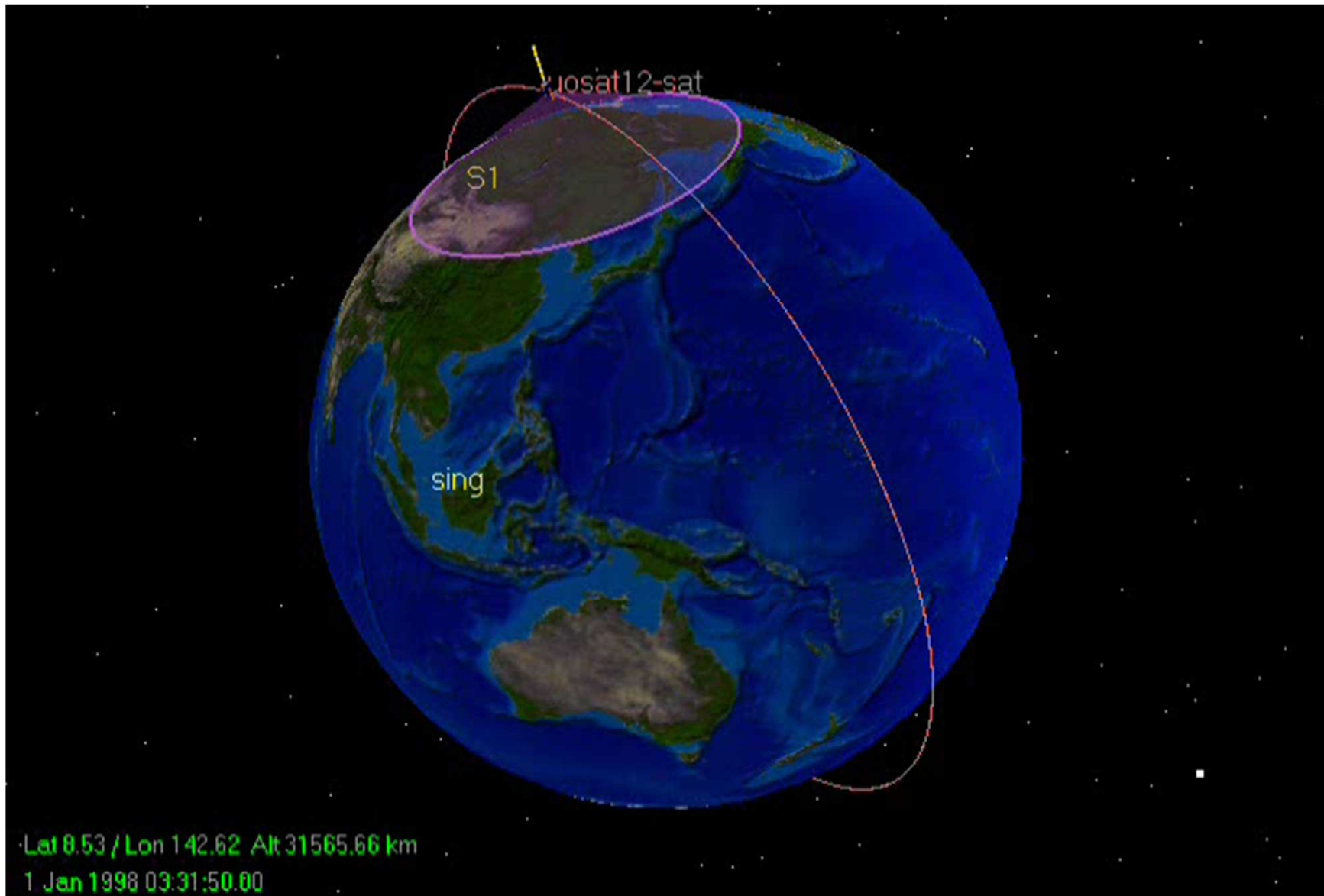
$$S = 2\pi R_E^2 (1 - \cos \theta)$$

The % of earth surface covered

$$S_P = \frac{2\pi R_E^2 (1 - \cos \theta)}{4\pi R_E^2} = \frac{(1 - \cos \theta)}{2}$$



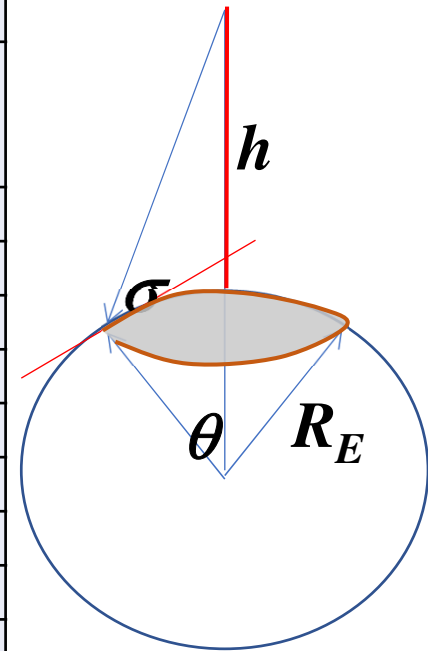
Single Satellite Coverage Animation



INITIAL CONSTELLATION DESIGN

Altitude

Altitude(h)(Km)	800		1000		1200	
Elevation (s)	Coverage Angle(θ)	% of Sphere Coverage(Sp)	Coverage Angle(θ)	% of Sphere Coverage(Sp)	Coverage Angle(θ)	% of Sphere Coverage(Sp)
0	27.3083	5.5725	30.1784	6.7768	32.6853	7.9175
10	18.9489	2.7096	21.6432	3.5251	24.0179	4.3291
20	13.3879	1.3588	15.6757	1.8597	17.7310	2.3752
30	9.6903	0.7134	11.5266	1.0084	13.2066	1.3224
40	7.1040	0.3838	8.5307	0.5532	9.8537	0.7376
50	5.1696	0.2034	6.2433	0.2965	7.2481	0.3995
60	3.6230	0.0999	4.3907	0.1467	5.1133	0.1990
70	2.3079	0.0406	2.8026	0.0598	3.2700	0.0814
80	1.1241	0.0096	1.3665	0.0142	1.5961	0.0194



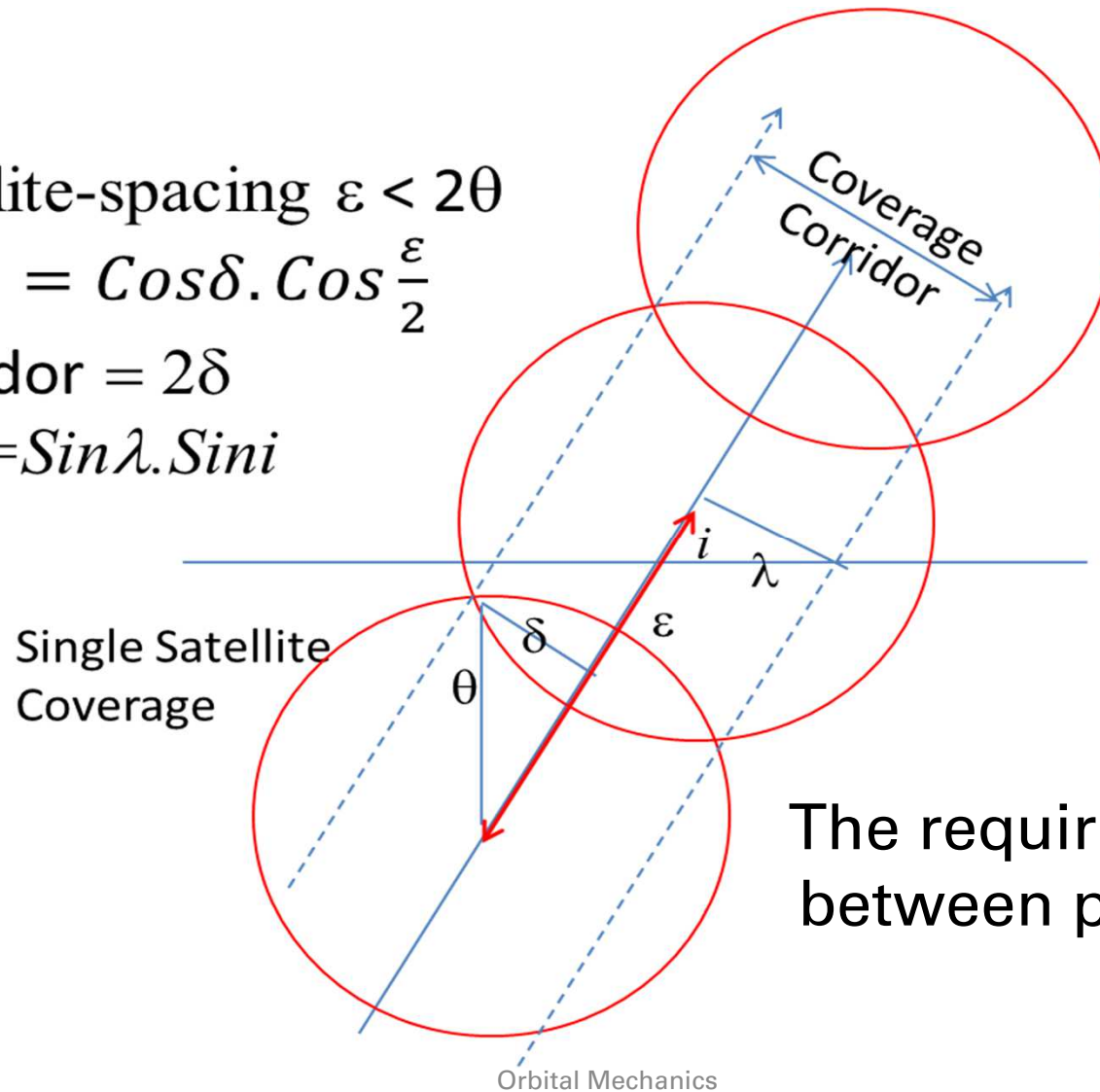
$$\theta = \cos^{-1} \left(\frac{\cos \sigma}{1 + \frac{h}{R_E}} \right) - \sigma$$

$$S_P = \frac{(1 - \cos \theta)}{2}$$

INITIAL CONSTELLATION DESIGN

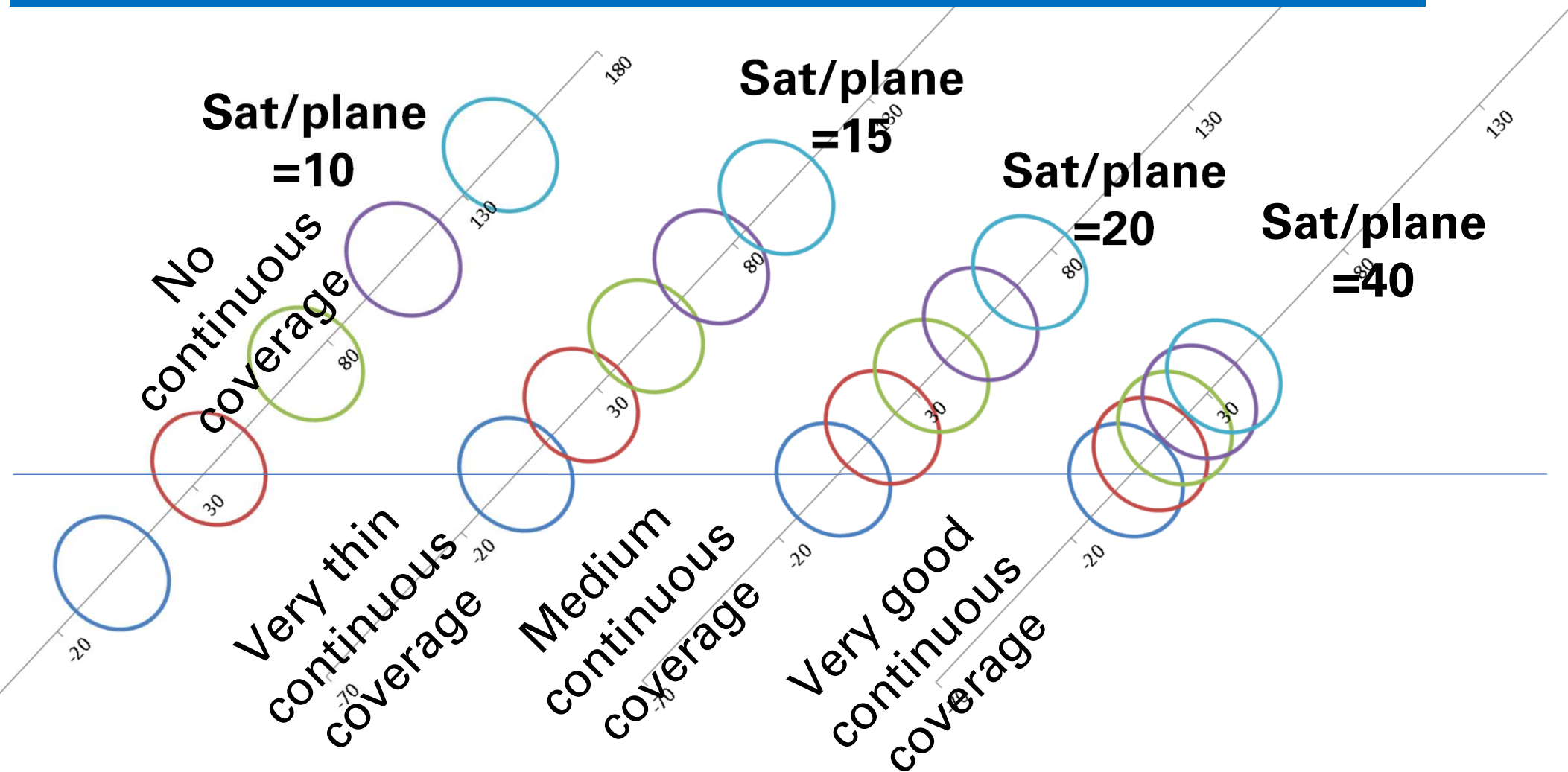
(Single Plane)

- Satellite-spacing $\varepsilon < 2\theta$
- $\cos\theta = \cos\delta \cdot \cos\frac{\varepsilon}{2}$
- Corridor = 2δ
- $\sin\delta = \sin\lambda \cdot \sin i$



The required spacing between planes $< 2\lambda$

No.of Sats Vs Coverage in a Single Plane



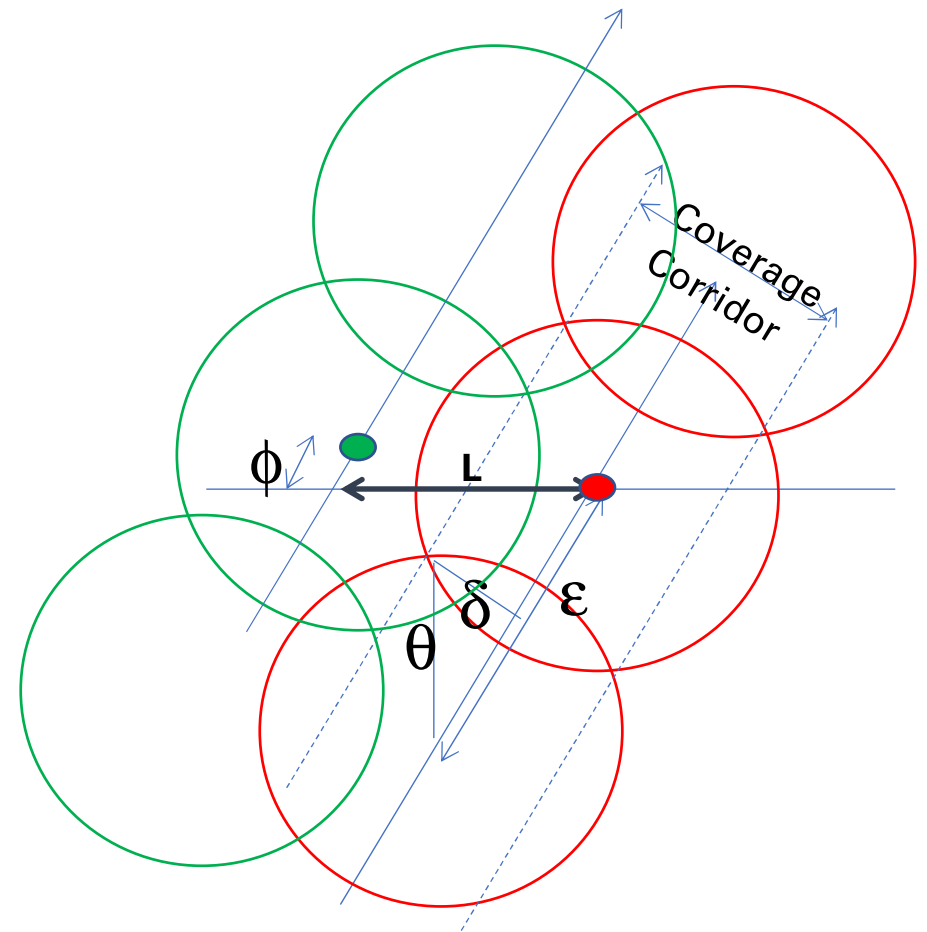
Note: Coverage circle Corresponds to a minimum elevation of 30 deg at 1200 km alt

Single Plane Constellation

[Single Plane Constellation Animation](#)

Multi Plane case

- $\varepsilon < 2\theta$
- $\cos\theta = \cos\delta \cdot \cos\frac{\varepsilon}{2}$
- Corridor = 2δ
- $\varepsilon = 2\pi/s$, $s = \text{Number of sats per plane}$
- $\sin\delta = \sin\lambda \sin i$
- Spacing between planes, $L < 2\lambda$



Coverage corridors from adjacent planes are just touching

(ϕ = Phasing in adjacent planes)

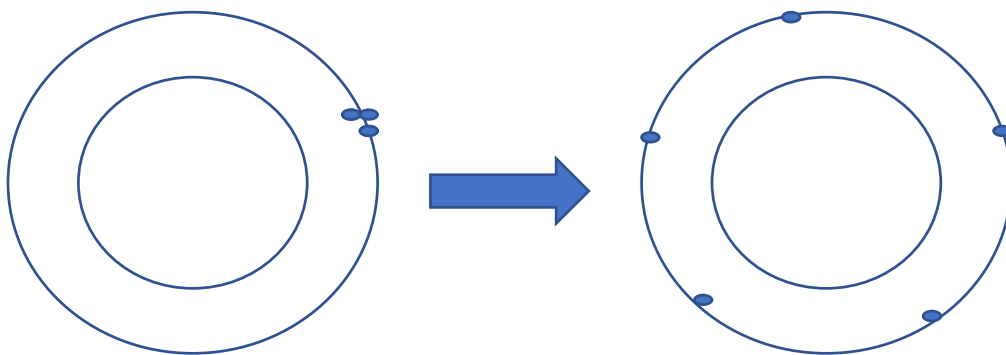
Impact of constellation parameters on services

Parameter	Impact on services
Altitude	Coverage radius, Orbit maintenance fuel budget, communication link margins,
Inclination	Latitude span for coverage. For HEO constellation 'Critical Inclination' is the only choice to maintain coverage pattern.
Satellites/plane	Width of coverage corridor, In-plane ISL characteristics
No.of planes	Distribution of snap shot coverage over the globe, Out-of-plane ISL characteristics.

Constellation Establishment

If ALL satellites of a plane (Q), is feasible in a single launch then,

- Launch Q satellites as a cluster into a desired orbital plane.
- Use in-plane phasing strategy to slowly shift the satellites to their respective orbital slots.
- Launch the next Q satellites as a cluster into another plane
- Perform in-plane phasing for the second plane.
- Repeat the process till all planes are established.



[ANIMATION](#)

Constellation maintenance and deorbiting

- Each satellite will experience common perturbations like aerodynamic, solar radiation, earth oblateness etc. due to same shape and mass.
- But due to attitude errors and other engineering factors (like mounting errors), there would be a differential portion for each perturbations.
- These two will require regular small maintenance maneuvers for each satellite.
- At EoL, a particular 'dead' satellite will need to be shifted to a higher orbit to make the 'slot' available for a replacement satellite.

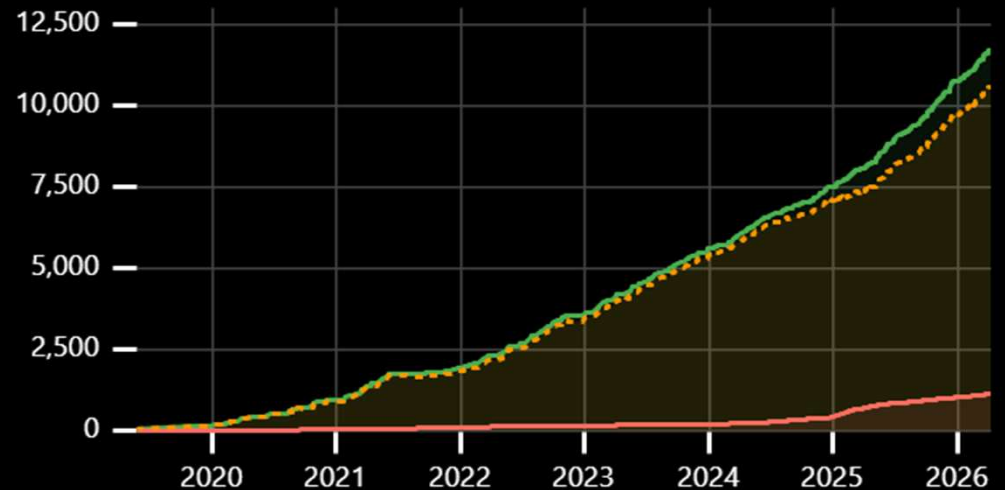
Starlink Constellation

- To provide high-speed, low-latency broadband internet service globally, particularly in underserved and rural areas.
- Thousands of small satellites in low Earth orbit at altitudes ranging from approximately 340 km to 550 km
- First launch in May 2019, Starlink has rapidly progressed with the launch of over 4,500 satellites as of early 2024.
- There are currently **10201 active starlink satellites** in orbit out of a planned constellation of 12,000

Recent Activity Summary

Period	Appeared	Decayed	Net Change
Today 2026-04-09	0	0	0
Week 2026-04-05	25 history	6 history	19
Month 2026-04-01	59 history	13 history	46
Year 2026-01-01	953 history	117 history	836

Constellation progress



Animations

[StarLinkAnimation_1](#)

[GPS Constellation Animation](#)

[Molnya Animation](#)

[**https://satellitemap.space/**](https://satellitemap.space/)

THANK YOU